PFP: Glyphs, rewrites, logits, and attention as checked proposal channels

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Code baseline: 59dde95a7860063c25d75db37f40af556e9f28ee

Abstract

PFP can accept candidates from natural-language prompts, glyph games, coloured-operad prompt decompositions, or model-internal logit and attention interventions. These channels can change which candidates are proposed while leaving the host's validity obligation fixed. We state and prove twelve finite results governing this architecture: (1) channel-independent host checking soundness, (2) exact prompt-to-cochain ℓ_1 primal-dual coboundary repair on directed cycles, (3) two-sided congruence-ideal absorption (**0** / `NullTree`) in the quotient coloured operad of prompt trees, (4) BPE prefix-automaton logit-mask viability with `F-null` token identity, (5) guarded Stage-1 slot retry with `NullTree` abstention (admission rate, conditional Stage-2 precision, and expected Stage-2 call savings), (6) the single-cycle partial-guard visibility gap showing that omitting the global cycle-sum equation leaves an infinite false-green kernel $\mathcal{K}_{\text{false}}(x) \cong \mathbb{Z} \setminus \{0\}$, (7) triangular chord-induced cycle decoupling on multi-cycle graphs ($\dim H_1(G) = b_1 \geq 2$) with a rank-$(|V|-1+b_1)$ partial-guard visibility gap $\mathcal{K}^{\text{C006}}_{\text{false}}(x) \cong \mathbb{Z}^{|V|-1+b_1} \setminus \{0\}$, (8) two-stage tree-then-chord `P*` factorization reducing maximum per-turn integer-slot arity on higher-Betti graphs ($b_1 \in \{4, \ldots, 8\}$) from $|V| - 1 + 2b_1$ to $\max(|V| - 1, b_1)$ while closing the two-stage false-green kernel, (9) elimination of the lazy false-refusal attractor on dichotomic exactness-vs-holonomy verification (`C005`) via a Fredholm-alternative Stage-1 guard G_{Fredholm}, (10) cohomological syndrome decoding $\mathbf{r}^* = S\mathbf{c}$ across `verdict`, single-edge `repair` (for both chord and spanning-tree perturbations), and `audit` modes on distractor-filtered 4-digit rank-difference graphs (`C001`), (11) elimination of tree-coboundary defect migration on retry via path-integrated B_T^{-1} guard diagnostics, and (12) independent exact certificate verification across all ten cohomology challenges (`C011`–`C020`, `b_cohomology`) closing the status-label false-`SOLVED` kernel. For historical context, we retain unverified summaries of six bounded paired C004 mechanism trials (including the held-out 40-case ladder `C004-S6` across $k \in \{3, \ldots, 12\}$), three bounded paired C006 multi-cycle trials (K_4 at $b_1 = 3$, higher-Betti $K_{3,3}, Q_3, K_5$, Petersen, W_7 at $b_1 \in \{4, 5, 6, 7\}$, and the held-out 60-board multi-cycle campaign `C006-S8` at $b_1 \in \{3, \ldots, 8\}$), one bounded paired C005 dichotomic Fredholm trial (`pub_cons_01..pub_cons_08`), one bounded paired C001 distractor-filtered syndrome trial (`12` public and adversarial poisoned cases across `verdict`, `repair`, and `audit`, $b_1 \in \{5, 7\}$), and one multi-stage `P*` cohomology certificate campaign (`B-COH-S1` across `C011`–`C020`) across a pinned local checkpoint (Qwen/Qwen2.5-0.5B-Instruct, rev 7ae557604adf67be50417f59c2c2f167def9a775) and accounted OpenRouter (openai/gpt-4o-mini and qwen/qwen-2.5-7b-instruct), together with an accounted Anthropic claude-fable-5-1 theorem-derivation receipt (msg_011Cg74HxZqarY627LKcRzQa).

The historical model-receipt files are unavailable in this revision. Their reported counts are not revalidated evidence. The finite examples and the new harness protocol have separate checks; the new paid campaign is NOT_RUN.

Mechanism inventory

Mechanism	Evidence
Mathematical statements (Theorems 1–3, Proposition 4, Theorems 5–12)	Written proofs under declared finite hypotheses + accounted claude-fable-5-1 derivation (fable_5_1_guard_theorem_receipt.json)
Finite example & theorem checks	Standard-library replay (code/replay.py) of glyph round trips, single-cycle and K_4 / higher-Betti ($b_1 \in \{4..8\}$) cochain duality, operad `NullTree` absorption, BPE prefix-automaton states, partial- vs. complete-guard kernel checks, dichotomic Fredholm-alternative guard closure, C001 cohomological syndrome decoding $\mathbf{r}^* = S\mathbf{c}$, path-integrated B_T^{-1} vs. edge-local B_T retry diagnostics, and `b_cohomology` (`C011`–`C020`) independent exact certificate verification
Bounded C001, C004, C005, C006 & `b_cohomology` (`C011`–`C020`) mechanism trials (Panel 1 local + Panel 2 OpenRouter + Anthropic claude-fable-5-1)	Thirteen committed JSON receipts replayed offline with zero network calls (c004_paired_benchmark_receipt.json, operad_cochain_ablation_receipt.json, p_star_operad_materialization_receipt.json, p_star_4slot_binding_receipt.json, p_star_verified_guard_receipt.json, fable_5_1_guard_theorem_receipt.json, c004_40case_hidden_p_star_receipt.json, c006_overlap_basis_p_star_receipt.json, c006_higher_betti_p_star_receipt.json, c006_60board_hidden_p_star_receipt.json, c005_dichotomic_fredholm_p_star_receipt.json, c001_distractor_cochain_p_star_receipt.json, b_cohomology_p_star_receipt.json)
Lean verification	No new formalization claimed in this draft
Authorization	None supplied

1. Proposal channels and the Three-Layer PFP host boundary

Fix a finite task P, a candidate language W, a solution relation $R(P, w)$, and a deterministic host checker C. A proposal channel μ is any distribution on raw messages produced by a prompt template, a multi-turn staged protocol, a local logit mask, or an external model provider. A separately supplied decoder maps well-formed messages to W and refuses malformed messages. The host replays the decoded witness against the original task P.

In the Omega Engine PFP runtime, this separation is organized into three non-collapsing layers:

1. **Layer 1 (Compositional Prompt Operad):** Typed prompt assembly and slot-grafting ($\circ_i$) over immutable source spans and congruence ideals.
2. **Layer 2 (Token-Frontier Grammar Projection):** BPE prefix-automaton logit masking (`gbnf_slots=k`) or strict response-schema projection restricting surface emission to a regular language $\mathcal{L}_k \subset \Sigma^*$.

3. **Layer 3 (Independent Host Cochain & Domain Oracles):** Exact rational cochain evaluation (Φ_1, Γ_2) and independent domain judges that certify or refuse the decoded candidate without consulting the proposal channel's internal scores.

2. Theorem 1: Channel-independent host checking soundness

Theorem 1 (Channel-Independent Soundness). Let $D : \Sigma^* \rightharpoonup W$ be a partial decoder that binds every admitted raw reply $m \in \Sigma^*$ to the fixed task P, and suppose the host checker $C : \{P\} \times W \to \{\mathsf{accept}, \mathsf{refuse}\}$ satisfies the soundness implication

$$C(P, w) = \mathsf{accept} \implies R(P, w).$$

Then for every proposal channel μ on Σ^*, the event of an accepted invalid candidate is empty:

$$\{m \in \Sigma^* : D(m) = w \text{ is defined}, \ C(P, w) = \mathsf{accept}, \ \neg R(P, w)\} = \emptyset,$$

and therefore has probability zero under μ. No acceptance-rate lower bound or search-completeness guarantee follows from soundness alone.

Proof. Fix any raw message $m \in \Sigma^*$ accepted by the host pipeline. By definition of admission, $w = D(m) \in W$ is defined and bound to P, and $C(P, w) = \mathsf{accept}$. By the soundness hypothesis on C, $R(P, w)$ holds. Because the deduction is pointwise in m and does not reference μ, no message m can simultaneously satisfy $C(P, D(m)) = \mathsf{accept}$ and $\neg R(P, D(m))$. Thus the bad set is empty and $\mu(\emptyset) = 0$ for every probability measure or empirical run μ. Conversely, a channel μ concentrating all mass on a single malformed string $m_0 \notin \mathrm{dom}(D)$ satisfies Theorem 1 vacuously with zero accepted solutions, demonstrating that validity and search success are independent obligations. □

3. Exact glyph transport

For the Boolean task $x + y = 1$, encode (x, y) by the string `<xy>`. The four domain values have four distinct strings: `<00>`, `<01>`, `<10>`, `<11>`. The decoder checks the brackets, length, and binary digits; it is inverse to the encoder on this declared domain. Hence checking a decoded string and checking its assignment are the same mathematical obligation. The strings use only a subset of A–Z, 0–9, and angle brackets.

More generally, a finite domain admits an injective finite-word code with a declared partial inverse. Proof is by indexing its elements and assigning distinct fixed-length binary words (with a separate empty-domain case). This is representation capacity. It does not show that an LLM can recover the desired word, solve every encoded question, or supply a missing domain interpretation. The typed prompt and required response matter because they fix that interpretation and the host's checking obligation.

4. Theorem 2: Exact prompt-to-cochain duality and single-cycle lightest-edge ℓ_1 projection

When a prompt task specifies local edge transitions or consistency claims across a finite 2-dimensional cell complex (V, E, F) over $\mathbb{Q}$, the host evaluates the candidate claim vector $c \in C^1 = \mathbb{Q}^E$ through the cochain complex

$$C^0 = \mathbb{Q}^V \xrightarrow{d_0} C^1 = \mathbb{Q}^E \xrightarrow{d_1} C^2 = \mathbb{Q}^F, \qquad d_1 \circ d_0 = 0,$$

where $(d_0 a)_{(u,v)} = a_v - a_u$ for node potentials $a \in \mathbb{Q}^V$, and $(d_1 c)_f = \sum_{e \in \partial f} \sigma_{f,e} c_e$ is the oriented cycle holonomy around 2-cell $f \in F$. Define the coboundary compatibility residual and the curvature residual by

$$\Phi_1(c) := \min_{a \in \mathbb{Q}^V} \|c - d_0 a\|_1, \qquad \Gamma_2(c) := \|d_1 c\|_1.$$

Theorem 2 (Single-Cycle Exact Coboundary Duality & Weighted ℓ_1 Repair). Let $G = (V, E, F)$ be a single directed k-cycle ($k \geq 3$) with vertices $v_0, \ldots, v_{k-1}$, directed edges $e_i = (v_i \to v_{(i+1) \bmod k})$, a single oriented k-gon face f_0, edge claims $c \in \mathbb{Q}^k$, positive edge weights $w \in \mathbb{Q}^k_{>0}$, and cycle holonomy sum $s := (d_1 c)_{f_0} = \sum_{i=0}^{k-1} c_i$. Then:

1. **Exactness and ℓ_1 Duality:** $\operatorname{im}(d_0) = \ker(d_1) = \{c' \in \mathbb{Q}^k : \sum_{i=0}^{k-1} c'_i = 0\}$, and
$$\Phi_1(c) = \Gamma_2(c) = |s| = \max_{z \in \mathbb{Q}^k : \partial_1 z = 0,\, \|z\|_\infty \leq 1} \langle z, c \rangle.$$
Whenever $\sum_{i=0}^{k-1} c'_i = 0$, the node potential $a \in \mathbb{Q}^k$ defined by $a_0 = 0$ and $a_m = \sum_{j=0}^{m-1} c'_j$ ($1 \leq m < k$) satisfies $d_0 a = c'$ identically.
2. **Weighted ℓ_1-Optimal Repair:** For any repaired cochain $c' = c + \Delta$ satisfying $d_1 c' = 0$ (equivalently $\sum_{i=0}^{k-1} \Delta_i = -s$), the weighted modification cost satisfies the lower bound
$$\|c' - c\|_{1,w} := \sum_{i=0}^{k-1} w_i |\Delta_i| \geq w_{\min} |s|, \qquad w_{\min} := \min_{0 \leq i < k} w_i.$$
Equality is attained by the single-edge coboundary repair at any lightest edge $i^* \in \arg\min_i w_i$:
$$c'_i = \begin{cases} c_{i^*} - s & \text{if } i = i^*, \\ c_i & \text{if } i \neq i^*, \end{cases}$$
and is certified with zero duality gap by the dual circulation witness $z^* = \operatorname{sgn}(s) w_{\min} \mathbf{1} \in \ker(\partial_1)$.

Proof. (1) Since $e_i = (v_i \to v_{(i+1) \bmod k})$, every telescopic sum around the cycle yields $\sum_{i=0}^{k-1} (d_0 a)_i = \sum_{i=0}^{k-1} (a_{(i+1) \bmod k} - a_i) = 0$, so $d_1 d_0 = 0$ and $\operatorname{im}(d_0) \subseteq \ker(d_1)$. Conversely, if $\sum_{i=0}^{k-1} c'_i = 0$, setting $a_0 = 0$ and $a_m = \sum_{j=0}^{m-1} c'_j$ gives $a_{m+1} - a_m = c'_m$ for $0 \leq m < k-1$ and $a_0 - a_{k-1} = -\sum_{j=0}^{k-2} c'_j = c'_{k-1}$, so $d_0 a = c'$. For any $a \in \mathbb{Q}^k$, the triangle inequality gives

$$\|c - d_0 a\|_1 = \sum_{i=0}^{k-1} |c_i - (d_0 a)_i| \geq \left| \sum_{i=0}^{k-1} (c_i - (d_0 a)_i) \right| = |s|,$$

and subtracting s from a single edge achieves $\|c - d_0 a\|_1 = |s|$ using the potential a of the resulting zero-sum vector. The dual cycle space $\ker(\partial_1) = \{z \in \mathbb{Q}^k : z_0 = z_1 = \cdots = z_{k-1}\}$ is spanned by $\mathbf{1}$, so $\max_{\partial_1 z = 0, \|z\|_\infty \leq 1} \langle z, c \rangle = \max_{\alpha \in [-1,1]} \alpha s = |s|$.

(2) For any $\Delta \in \mathbb{Q}^k$ with $\sum_{i=0}^{k-1} \Delta_i = -s$,

$$\sum_{i=0}^{k-1} w_i |\Delta_i| \geq w_{\min} \sum_{i=0}^{k-1} |\Delta_i| \geq w_{\min} \left| \sum_{i=0}^{k-1} \Delta_i \right| = w_{\min} |s|.$$

Setting $\Delta_{i^*} = -s$ and $\Delta_j = 0$ ($j \neq i^*$) satisfies $\sum_i (c_i + \Delta_i) = s - s = 0$ and achieves $\sum_i w_i |\Delta_i| = w_{i^*} |s| = w_{\min} |s|$. $\square$

5. Theorem 3 & Proposition 4: Coloured operad ideal absorption and BPE prefix-automaton projection

5.1 Theorem 3: Coloured operad quotient and two-sided `NullTree` absorption

Let $\mathcal{C}$ be the finite set of prompt semantic colours (`RawTextSpan`, IsolatedVariable, `Equation`, SubstitutionStep, RationalValue, InvariantWitness, `TargetResult`), and let $\mathcal{O}$ be the free $\mathcal{C}$-coloured operad of syntax trees generated by typed n-ary operators $f \in \mathcal{O}^+(c_1, \dots, c_n; c_{\text{out}})$ and leaf spans $[i, j]$ extracted from a source document D. Let $\mathcal{I} \subset \mathcal{O}$ be the subset of trees containing any of the following defects: (i) a non-exact or boolean-coerced numeric value (IDEAL_RATIONAL_DOMAIN), (ii) a leaf span whose content or SHA-256 digest does not match $D[i : j]$ (IDEAL_SPAN_IMMUTABILITY), (iii) a slot-grafting operation $f \circ_m g$ whose child output colour $\text{out}(g)$ does not equal the m-th input colour $\text{in}_m(f)$ or whose slot index $m \notin \{0, \dots, \text{arity}(f) - 1\}$, or (iv) a candidate binding neutralized by an active AnnihilatorDecoy witness ($\mathcal{O}^-$).

Theorem 3 (Two-Sided Ideal Absorption in $\mathcal{O}/\mathcal{I}$). The subset $\mathcal{I}$ is a two-sided operad ideal: for every tree $t \in \mathcal{I}$ and every tree $u \in \mathcal{O}$, both $u \circ_m t \in \mathcal{I}$ and $t \circ_m u \in \mathcal{I}$. Consequently, the quotient coloured operad $\mathcal{O}/\mathcal{I} \coloneqq (\mathcal{O} \setminus \mathcal{I}) \sqcup \{\mathbf{0}\}$ with distinguished null element $\mathbf{0}$ (`NullTree`) satisfies the two-sided absorption law

$$f \circ_m \mathbf{0} = \mathbf{0}, \qquad \mathbf{0} \circ_m g = \mathbf{0},$$

and any composite tree $T \in \mathcal{O}/\mathcal{I}$ satisfies $T \neq \mathbf{0}$ if and only if every leaf span is byte-identical to its declared slice in D, every parameter belongs to $\mathbb{Q}$, every internal graft matches its declared colour signature, and no active decoy witness is triggered.

Proof. By induction on tree depth. A single leaf or operator violation places the local node in $\mathcal{I}$, which projects to $\mathbf{0} \in \mathcal{O}/\mathcal{I}$. Because containing a defect in $\mathcal{I}$ is hereditary under subtree inclusion—if a subtree t contains a malformed span, non-rational parameter, or colour mismatch, then any larger tree $u \circ_m t$ or $t \circ_m u$ containing t also contains that defect—$\mathcal{I}$ is closed under both left and right partial composition $\circ_m$. Collapsing $\mathcal{I}$ to a single point $\mathbf{0}$ therefore makes $\circ_m$ well-defined on $\mathcal{O}/\mathcal{I}$ with $f \circ_m \mathbf{0} = \mathbf{0}$ and $\mathbf{0} \circ_m g = \mathbf{0}$. □

5.2 Proposition 4: BPE prefix-automaton logit mask viability and `F-null` identity

Let $\mathcal{L}_k \subset \Sigma^*$ be the regular language of k space-separated signed ASCII integers (integer_slots_gbnf(k)):

$$\mathcal{L}_k = \left\{ w_1 \,\|\, \texttt{" "} \,\|\, w_2 \,\|\, \dots \,\|\, \texttt{" "} \,\|\, w_k \;\middle|\; w_j \in \{\texttt{"0"}\} \cup (\{\varepsilon, \texttt{"-"}\} \cdot \{\texttt{1..9}\} \cdot \{\texttt{0..9}\}^*) \right\},$$

and let $\text{Pref}(\mathcal{L}_k) = \{u \in \Sigma^* : \exists v \in \Sigma^*,\ u \,\|\, v \in \mathcal{L}_k\}$ be its set of viable prefixes, recognized by the six-component prefix-state DFA $\delta_{\text{DFA}}(u) = (\text{viable}, \text{complete}, \text{slots_seen}, \text{in_number}, \text{after_minus}, \text{after_zero})$.

Proposition 4 (BPE Token-Frontier Projection & `F-null` Identity). For any BPE tokenizer vocabulary $V \subset \Sigma^+$ with end-of-sequence token $\tau_{\text{eos}} \notin V$ and any viable prefix $S_t \in \text{Pref}(\mathcal{L}_k)$ at generation step t, define the admissible token frontier

$$V_{\text{valid}}(S_t) \coloneqq \{\tau \in V : S_t \,\|\, \tau \in \text{Pref}(\mathcal{L}_k)\} \cup \{\tau_{\text{eos}} : S_t \in \mathcal{L}_k\},$$

and define the masked logit vector $\tilde{\ell}_v^{(t)} = \ell_v^{(t)}$ if $v \in V_{\text{valid}}(S_t)$ and $-\infty$ otherwise. Then:

1. **Prefix Viability & Complete-Word Soundness:** If $S_0 = \varepsilon$ and at each step $t < T$ the sampled token τ_t has finite masked logit ($\tau_t \in V_{\text{valid}}(S_t)$), then $S_t \in \text{Pref}(\mathcal{L}_k)$ for all $t \geq 0$, and

if generation terminates via $\tau_T = \tau_{\text{eos}}$, then $S_T \in \mathcal{L}_k$ (`parse_integer_slots(S_T, k)` succeeds).

2. **BPE Multi-Character Boundary Completeness:** A merged BPE token $\tau \in V$ spanning a slot boundary (such as $\tau =$ `" -"` or `" 12"`) belongs to $V_{\text{valid}}(S_t)$ whenever S_t ends a valid integer slot with slots_seen$(S_t) < k$, because $\delta_{\text{DFA}}(S_t \| \tau)$ transitions across the space character and opens slot slots_seen$(S_t) + 1$.
3. `F-null` **Token Identity:** When the logit processor is instantiated with the full vocabulary frontier $V^{\text{null}}_{\text{valid}}(S_t) = V \cup \{\tau_{\text{eos}}\}$, $\tilde{\ell}^{(t)} = \ell^{(t)}$ at every step t, so greedy or fixed-seed decoding produces a token sequence identical to unhooked decoding.

Proof. (1) follows by induction on t: $S_0 = \varepsilon \in \text{Pref}(\mathcal{L}_k)$, and $\tau_t \in V_{\text{valid}}(S_t) \setminus \{\tau_{\text{eos}}\}$ implies $S_{t+1} = S_t \,\|\, \tau_t \in \text{Pref}(\mathcal{L}_k)$ by definition of $V_{\text{valid}}(S_t)$, while $\tau_{\text{eos}} \in V_{\text{valid}}(S_T)$ holds if and only if $S_T \in \mathcal{L}_k$. (2) follows because δ_{DFA} advances character-by-character over the concatenation $S_t \,\|\, \tau$: after a completed slot $m < k$, `' '` transitions to (slots_seen $= m$, in_number $=$ false), and `'-'` then transitions to (slots_seen $= m+1$, in_number $=$ true, after_minus $=$ true), which is a viable prefix of $\mathcal{L}_k$. (3) is immediate because adding 0 to every coordinate of $\ell^{(t)}$ leaves $\arg\max_v \ell^{(t)}_v$ and softmax$(\ell^{(t)})$ invariant. □

5.3 Theorem 5: Guarded Stage-1 slot retry with `NullTree` abstention

Consider a 2-stage `P*` operad pipeline $X \xrightarrow{f_1} Z_1 \xrightarrow{f_2 \circ_i z_1} Y$ where Stage 1 proposes a 4-slot invariant tuple $z_1 = (\hat{s}, \hat{i}^*, \hat{c}_{i^*}, \hat{c}'_{i^*}) \in \mathbb{Z}^4$ for a directed k-cycle $x = (c, w)$ with unique canonical truth tuple

$$z_1^*(x) = (s(c),\ i^*(w),\ c_{i^*},\ c_{i^*} - s(c)), \qquad i^*(w) := \min \arg\min_{0 \le j < k} w_j.$$

Let $G_1 : \mathcal{X} \times \mathbb{Z}^4 \to \{0, 1\}$ be a deterministic Stage-1 host guard, let $\pi_{\mathcal{J}}$ project any rejected Stage-1 output ($G_1(x, z_1) = 0$) to `NullTree` ($\mathbf{0}$), and let $R \ge 1$ be a bounded Stage-1 retry cap (1 initial attempt + at most $R - 1$ diagnostic retries).

Theorem 5 (Guarded Stage-1 Slot Retry with Ideal Abstention — derived with claude-fable-5-1, msg_011Cg74HxZqarY627LKcRzQa). Suppose the Stage-1 guard G_1 is sound and complete for $z_1^*(x)$, meaning $G_1(x, z_1) = 1 \iff z_1 = z_1^*(x)$. Let $p_1(x) := \mathbb{P}(Z_1^{(1)} = z_1^*(x) \mid x)$ be the Attempt-1 exact Stage-1 probability, and under the conditional independence assumption that each diagnostic retry $r \in \{2, \dots, R\}$ given prior failure succeeds with probability $p_{1,\text{retry}}(x) := \mathbb{P}(Z_1^{(r)} = z_1^*(x) \mid x, G_1(x, Z_1^{(r-1)}) = 0)$, let $p_2(y^* \mid x, z_1)$ denote the Stage-2 probability of emitting the exact ℓ_1-optimal cochain repair $y^*(x)$ when z_1 is grafted into f_2. Then:

1. **Stage-1 Admission Probability:** The probability that Stage 1 is admitted to Stage 2 within R bounded attempts is
$$q_{G_1,R}(x) = 1 - (1 - p_1(x))(1 - p_{1,\text{retry}}(x))^{R-1}.$$
2. **Admission-Conditional Stage-2 Precision:** Because $G_1(x, z_1) = 1 \iff z_1 = z_1^*(x)$ and $\pi_{\mathcal{J}}$ absorbs all $G_1 = 0$ terminations into `NullTree` ($f_2 \circ_i \mathbf{0} = \mathbf{0}$ by Theorem 3), conditioning on Stage-2 admission eliminates all corrupted Stage-1 grafts:
$$\mathbb{P}_{\text{ver},R}(Y = y^*(x) \mid x, \text{Stage 2 admitted}) = p_2(y^*(x) \mid x,\ z_1^*(x)).$$
3. **Unconditional Exact Repair Rate vs. Unverified Grafting:** Under unverified 1-shot grafting ($R = 1$, no guard), the unconditional optimal repair rate is
$$\mathbb{P}_{\text{unv}}(Y = y^*(x) \mid x) = p_1(x)\, p_2(y^*(x) \mid x, z_1^*(x)) + \sum_{z_1 \ne z_1^*(x)} \mathbb{P}(Z_1 = z_1 \mid x)\, p_2(y^*(x) \mid x, z_1),$$
whereas under guarded retry with `NullTree` abstention,
$$\mathbb{P}_{\text{ver},R}(Y = y^*(x) \mid x) = q_{G_1,R}(x)\, p_2(y^*(x) \mid x, z_1^*(x)).$$

In particular, whenever $p_2(y^*(x) \mid x, z_1) = 0$ for all corrupted tuples $z_1 \neq z_1^*(x)$ (faithful Stage-2 slot substitution), guarded retry improves unconditional optimal repair by the exact gain

$$\mathbb{P}_{\mathrm{ver},R}(Y = y^*(x) \mid x) - \mathbb{P}_{\mathrm{unv}}(Y = y^*(x) \mid x) = (1 - p_1(x))\Big[1 - (1 - p_{1,\mathrm{retry}}(x))^{R-1}\Big]\, p_2(y^*(x) \mid x, z_1^*(x)) \geq 0.$$

4. **Expected Stage-2 Call Savings:** If a Stage-2 call costs $c_2 > 0$, `NullTree` abstention on persistent Stage-1 guard failure saves
$$\mathbb{E}[\Delta C_2 \mid x] = (1 - q_{G_1,R}(x))\, c_2$$
in expected Stage-2 expenditure compared to unverified Stage-2 dispatch.

Proof. (1) Stage 1 fails admission across all R attempts if and only if Attempt 1 fails $(1 - p_1(x))$ and each of the $R - 1$ retries fails $((1 - p_{1,\mathrm{retry}}(x))^{R-1})$; taking the complement gives $q_{G_1,R}(x)$. (2) On the event of Stage-2 admission, the admitted tuple $\hat{z}_1$ satisfies $G_1(x, \hat{z}_1) = 1$, which by soundness and completeness of G_1 implies $\hat{z}_1 = z_1^*(x)$ deterministically; thus the Stage-2 law conditioned on admission is $p_2(\cdot \mid x, z_1^*(x))$. (3) follows by the law of total probability over $z_1 \in \mathbb{Z}^4$ and substituting $q_{G_1,R}(x) - p_1(x) = (1 - p_1(x))[1 - (1 - p_{1,\mathrm{retry}}(x))^{R-1}]$. (4) Unverified grafting dispatches Stage 2 with probability 1, whereas guarded grafting dispatches Stage 2 with probability $q_{G_1,R}(x)$ and projects the remaining $1 - q_{G_1,R}(x)$ mass to `NullTree` at zero Stage-2 cost. □

5.4 Theorem 6: Partial-guard visibility gap and false-green kernel $\mathcal{K}_{\mathbf{false}}(x) \cong \mathbb{Z} \setminus \{0\}$

Suppose a Stage-1 guard checks only the **local** slot invariants on $z_1 = (\hat{s}, \hat{i}^*, \hat{c}_{i^*}, \hat{c}'_{i^*}) \in \mathbb{Z}^4$—namely canonical minimum-weight index selection, exact edge-claim lookup, and the linear subtraction relation—without checking the global cycle-sum equation $\hat{s} = \sum_{j=0}^{k-1} c_j$:

$$G_{\mathrm{partial}}(x, z_1) = 1 \iff \Big(\hat{i}^* = i^*(w)\Big) \wedge \Big(\hat{c}_{i^*} = c_{\hat{i}^*}\Big) \wedge \Big(\hat{c}'_{i^*} + \hat{s} = \hat{c}_{i^*}\Big),$$

whereas the **complete** guard adjoins the global cycle-sum check:

$$G_{\mathrm{complete}}(x, z_1) = 1 \iff G_{\mathrm{partial}}(x, z_1) = 1 \wedge \left(\hat{s} = \sum_{j=0}^{k-1} c_j\right).$$

Theorem 6 (Partial-Guard Visibility Gap & False-Green Kernel — derived with claude-fable-5-1, msg_011Cg74HxZqarY627LKcRzQa). For every directed k-cycle $x = (c, w)$:

1. **Infinite False-Green Kernel under $G_{\mathbf{partial}}$:** The set of false-green Stage-1 tuples admitted by G_{partial} is
$$\mathcal{K}_{\mathrm{false}}(x) := \big\{z_1 \in \mathbb{Z}^4 : G_{\mathrm{partial}}(x, z_1) = 1 \wedge z_1 \neq z_1^*(x)\big\} \cong \mathbb{Z} \setminus \{0\},$$
parameterized bijectively by the non-zero cycle-sum error $\Delta s := \hat{s} - s(c) \in \mathbb{Z} \setminus \{0\}$ via
$$z_1(\Delta s) = (s(c) + \Delta s,\ i^*(w),\ c_{i^*},\ c_{i^*} - s(c) - \Delta s)\,.$$
2. **Induced Stage-2 Cochain Residual (Local Green / Global Red):** If Stage 2 substitutes a false-green tuple $z_1(\Delta s) \in \mathcal{K}_{\mathrm{false}}(x)$ into edge $i^* = i^*(w)$ while preserving $c'_j = c_j$ for $j \neq i^*$, the resulting Stage-2 edge cochain $c'(\Delta s)$ has non-zero cycle holonomy
$$\sum_{j=0}^{k-1} c'_j(\Delta s) = -\Delta s \neq 0, \qquad \Gamma_2(c'(\Delta s)) = |\Delta s| > 0, \qquad \Phi_{1,w}(c'(\Delta s)) = w_{i^*}|\Delta s| > 0,$$
and is therefore rejected by the Layer-3 cochain verifier as NonClosedEdgeData / ClosedNo nExactClass.
3. **Kernel Closure under $G_{\mathbf{complete}}$:** Adjoining $\hat{s} = \sum_{j=0}^{k-1} c_j$ forces $\Delta s = 0$, collapsing the false-green kernel to the empty set ($\mathcal{K}_{\mathrm{false}}^{\mathrm{complete}}(x) = \emptyset$) and making $G_{\mathrm{complete}}(x, z_1) = 1 \iff$

$z_1 = z_1^*(x)$.

Proof. (1) Any $z_1 = (\hat{s}, \hat{i}^*, \hat{c}_{i^*}, \hat{c}'_{i^*}) \in \mathbb{Z}^4$ satisfying $G_{\text{partial}}(x, z_1) = 1$ has $\hat{i}^* = i^*(w)$ and $\hat{c}_{i^*} = c_{i^*}$ uniquely fixed, while $\hat{c}'_{i^*} = c_{i^*} - \hat{s}$ is uniquely determined by $\hat{s} \in \mathbb{Z}$. Setting $\Delta s = \hat{s} - s(c) \in \mathbb{Z}$ gives the bijection $\Delta s \mapsto z_1(\Delta s)$, with $\Delta s = 0 \iff z_1 = z_1^*(x)$. (2) Substituting $c'_{i^*} = c_{i^*} - s(c) - \Delta s$ into c yields cycle sum $\sum_{j \neq i^*} c_j + (c_{i^*} - s(c) - \Delta s) = s(c) - s(c) - \Delta s = -\Delta s \neq 0$, so Theorem 2 gives $\Gamma_2(c'(\Delta s)) = |\Delta s| > 0$ and $\Phi_{1,w}(c'(\Delta s)) = w_{i^*}|\Delta s| > 0$. (3) Adding $\hat{s} = s(c)$ forces $\Delta s = 0$. □

5.5 Theorem 7: Triangular chord-induced cycle decoupling and multi-cycle partial-guard visibility gap $\mathcal{K}^{\mathbf{C006}}_{\mathbf{false}}(x) \cong \mathbb{Z}^{|V|-1+b_1} \setminus \{0\}$

Let $G = (V, E)$ be a finite connected directed graph with $|V|$ vertices, $|E|$ edges, spanning tree $T \subset E$ ($|T| = |V| - 1$) rooted at $v_0 \in V$, and first Betti number $b_1 := \dim H_1(G; \mathbb{Q}) = |E \setminus T| \geq 2$ chords $g_1, \ldots, g_{b_1} \in E \setminus T$, where $g_m = (u_m \to v_m)$. Each chord g_m closes a unique chord-induced oriented cycle γ_m whose support lies in $T \cup \{g_m\}$ with $+1$ orientation on g_m and signed tree path $-\pi_T(u_m \to v_m)$, even when multiple cycles $\gamma_1, \ldots, \gamma_{b_1}$ share high-incidence tree edges $e_{\text{shared}} \in T$ (for example, on K_4 with a non-star spanning tree T, $|V| = 4, |E| = 6, b_1 = 3$, and a shared tree edge traversed by all 3 chord-induced cycles).

For any edge claim vector $c \in \mathbb{Z}^E$, define the unique tree-integrated vertex potential $p_T^* \in \mathbb{Z}^V$ by $p_T^*(v_0) = 0$ and $p_T^*(v) - p_T^*(u) = c_{(u,v)}$ for every tree edge $(u, v) \in T$, and define the canonical Stage-1 ($|V| - 1 + 2b_1$)-slot tuple (9 slots on K_4) in causal dependency order:

$$z_1^*(x) := \Big((p_T^*(v_j))_{j=1}^{|V|-1},\ (c'^*_{g_m})_{m=1}^{b_1},\ (r_m^*)_{m=1}^{b_1} \Big) \in \mathbb{Z}^{|V|-1+2b_1},$$

where $c'^*_{g_m} := p_T^*(v_m) - p_T^*(u_m)$ and $r_m^* := \sum_{e \in \gamma_m} \sigma_{m,e} c_e = c_{g_m} - c'^*_{g_m}$. Compare the **partial chord-subtraction guard**

$$G^{\text{C006}}_{\text{partial}}(x, z_1) = 1 \iff \bigwedge_{m=1}^{b_1} \left(\hat{c}'_{g_m} + \hat{r}_m = c_{g_m} \right)$$

against the **complete triangular-basis guard**

$$G^{\text{C006}}_{\text{complete}}(x, z_1) = 1 \iff \bigwedge_{(u,v) \in T} (\hat{p}(v) - \hat{p}(u) = c_{(u,v)}) \ \wedge \ \bigwedge_{m=1}^{b_1} (\hat{c}'_{g_m} = \hat{p}(v_m) - \hat{p}(u_m)) \ \wedge \ \bigwedge_{m=1}^{b_1} (\hat{c}'_{g_m} + \hat{r}_m = c_{g_m}).$$

Theorem 7 (Triangular Chord-Induced Cycle Decoupling & Multi-Cycle Visibility Gap). For every connected graph $G = (V, E)$ with spanning tree T and $b_1 = |E \setminus T| \geq 2$ chords:

1. **Exact Triangular Chord Decoupling:** The cycle-chord incidence matrix of the chord-induced cycle basis $\{\gamma_1, \ldots, \gamma_{b_1}\}$ restricted to the chord coordinates $\{g_1, \ldots, g_{b_1}\} = E \setminus T$ is the $b_1 \times b_1$ identity matrix I_{b_1} ($\sigma_{m,g_\ell} = \delta_{m,\ell}$). Consequently, regardless of how many cycles γ_m overlap on shared spanning-tree edges $e \in T$, fixing the tree claims $c'_e = c_e$ ($e \in T$) and setting $c'_{g_m} = c'^*_{g_m} = p_T^*(v_m) - p_T^*(u_m) = c_{g_m} - r_m^*$ for each chord $m \in \{1, \ldots, b_1\}$ yields an exact coboundary $c' = d_0 p_T^* \in \text{im}(d_0)$ with zero defect across all cycles simultaneously ($\Phi_{1,w}(c') = 0$ and $\Gamma_2(c') = 0$). Moreover, if each tree edge weight satisfies $w_e \geq \sum_{m : e \in \gamma_m} w_{g_m}$ (the tree-dominant weight regime), then c' is a global minimum-weight ℓ_1 coboundary repair of c with $\|c' - c\|_{1,w} = \sum_{m=1}^{b_1} w_{g_m} |r_m^*|$.

2. **Rank-$(|V|-1+b_1)$ Partial-Guard Visibility Gap:** The false-green kernel of tuples passing $G^{\mathrm{C006}}_{\mathrm{partial}}$ while failing $G^{\mathrm{C006}}_{\mathrm{complete}}$ is canonically isomorphic to the punctured free $\mathbb{Z}$-module of rank $|V|-1+b_1$ $(= 2b_1 = 6$ on $K_4)$:
$$\mathcal{K}^{\mathrm{C006}}_{\mathrm{false}}(x) := \left\{ z_1 \in \mathbb{Z}^{|V|-1+2b_1} : G^{\mathrm{C006}}_{\mathrm{partial}}(x, z_1) = 1 \ \wedge \ z_1 \neq z_1^*(x) \right\} \cong \mathbb{Z}^{|V|-1+b_1} \setminus \{0\},$$
parameterized bijectively by $(\Delta p, \Delta r) \in \mathbb{Z}^{|V|-1} \times \mathbb{Z}^{b_1} \setminus \{(0,0)\}$ via
$$\hat{p}(v_j) = p_T^*(v_j) + \Delta p_j, \qquad \hat{r}_m = r_m^* + \Delta r_m, \qquad \hat{c}'_{g_m} = c_{g_m} - r_m^* - \Delta r_m = c'^*_{g_m} - \Delta r_m.$$
In particular, any non-zero chord-residual error $\Delta r \neq 0$ induces non-zero post-repair cycle holonomy $(d_1 c'(\Delta r))_{\gamma_m} = -\Delta r_m$ $(\Gamma_2(c'(\Delta r)) = \|\Delta r\|_1 > 0)$, while any non-zero potential error $\Delta p \neq 0$ with $\Delta r = 0$ causes Stage-2 `VERIFY` to fail on the tree-edge coboundary check $\hat{p}(v) - \hat{p}(u) = c'_{(u,v)}$.
3. **Complete-Guard Closure:** $G^{\mathrm{C006}}_{\mathrm{complete}}(x, z_1) = 1 \iff z_1 = z_1^*(x)$, collapsing $\mathcal{K}^{\mathrm{C006}}_{\mathrm{false}}$ to $\emptyset$ and ensuring that every Stage-2 admitted session plan (`ASSERT` $\to$ `CUT` $\to$ `VERIFY`) carries an exact zero-defect coboundary $c' = d_0 \hat{p}$ and its matching vertex potential $\hat{p} = p_T^*$.

Proof. (1) By construction of the spanning-tree cycle basis, chord g_ℓ belongs to γ_m if and only if $\ell = m$. Since p_T^* integrates c along the unique path in T from v_0, for every tree edge $e = (u, v) \in T$ we have $c'_e = c_e = p_T^*(v) - p_T^*(u)$, and for every chord $g_m = (u_m, v_m) \in E \setminus T$ we have $c'_{g_m} = p_T^*(v_m) - p_T^*(u_m)$. Thus $c' = d_0 p_T^*$, so $\Phi_{1,w}(c') = 0$ and $\Gamma_2(c') = 0$. For any alternative potential $p = p_T^* + \delta p$, $\|c - d_0 p\|_{1,w} = \sum_{e \in T} w_e |(d_0 \delta p)_e| + \sum_{m=1}^{b_1} w_{g_m} |r_m^* - (d_0 \delta p)_{g_m}|$. Expanding $(d_0 \delta p)_{g_m} = \sum_{e \in \gamma_m \cap T} \sigma_{m,e} (d_0 \delta p)_e$ and applying the triangle inequality shows that whenever $w_e \geq \sum_{m : e \in \gamma_m} w_{g_m}$ for all $e \in T$, setting $d_0 \delta p = 0$ on T ($\delta p = \mathrm{const}$) achieves the global minimum $\sum_{m=1}^{b_1} w_{g_m} |r_m^*|$. (2) Under $G^{\mathrm{C006}}_{\mathrm{partial}}$, the $|V| - 1$ tree-potential slots $\hat{p} = p_T^* + \Delta p \in \mathbb{Z}^{|V|-1}$ and the b_1 residual slots $\hat{r} = r^* + \Delta r \in \mathbb{Z}^{b_1}$ are arbitrary integers, and $\hat{c}'_{g_m} = c_{g_m} - \hat{r}_m = c'^*_{g_m} - \Delta r_m$ is uniquely determined by $\hat{r}_m$. Thus $(\Delta p, \Delta r) \mapsto z_1(\Delta p, \Delta r)$ is a bijection from $\mathbb{Z}^{|V|-1+b_1} \setminus \{0\}$ onto $\mathcal{K}^{\mathrm{C006}}_{\mathrm{false}}(x)$. Around cycle γ_m, the repaired cochain $c'(\Delta r)$ sums to $(p_T^*(u_m) - p_T^*(v_m)) + c'_{g_m} = -c'^*_{g_m} + (c'^*_{g_m} - \Delta r_m) = -\Delta r_m$. (3) Under $G^{\mathrm{C006}}_{\mathrm{complete}}$, tree-edge integration forces $\hat{p}(v_j) = p_T^*(v_j)$ $(\Delta p = 0)$ by induction along T from $p(v_0) = 0$; chord potential difference then forces $\hat{c}'_{g_m} = \hat{p}(v_m) - \hat{p}(u_m) = c'^*_{g_m}$; and chord subtraction forces $\hat{r}_m = c_{g_m} - \hat{c}'_{g_m} = r_m^*$ $(\Delta r = 0)$. □

5.6 Theorem 8: Two-stage tree-then-chord `P*` factorization and slot-arity reduction on higher-Betti graphs ($b_1 \in \{4, \ldots, 7\}$)

On higher-Betti graphs ($b_1 = |E \setminus T| \geq 4$, such as $K_{3,3}$ with $b_1 = 4$, Q_3 with $b_1 = 5$, K_5 and Petersen with $b_1 = 6$, and W_7 with $b_1 = 7$), single-stage triangular basis materialization requires emitting $A_{\text{1-stage}}(G) := (|V|-1) + 2b_1 \in \{13, 17, 16, 21, 21\}$ signed integer slots in a single autoregressive turn while tracking multi-hop spanning-tree paths and chord subtractions simultaneously.

Instead, we factorize the exact sequence

$$0 \longrightarrow C^0_{\mathrm{root}=0}(G; \mathbb{Z}) \xrightarrow[\cong]{d_0|_T} C^1(T; \mathbb{Z}) \longrightarrow 0, \qquad C^1(G; \mathbb{Z}) = C^1(T; \mathbb{Z}) \oplus C^1(E \setminus T; \mathbb{Z}),$$

into two sequential guarded `P*` Stage-1 cuts before Stage 2: - **Stage 1a (Spanning-Tree Potential Materialization, slot arity $A_{1a} = |V| - 1$):** Proposes $\hat{p}_T = (\hat{p}(v_1), \ldots, \hat{p}(v_{|V|-1})) \in \mathbb{Z}^{|V|-1}$ under `gbnf_slots = |V|-1`, checked by the deterministic **tree-integration guard**

$$G_T(x, \hat{p}_T) = 1 \iff \bigwedge_{e=(u,v) \in T} (\hat{p}(v) - \hat{p}(u) = c_e), \qquad \hat{p}(v_0) := 0.$$

If $G_T = 0$ after at most $R_{1a} \geq 1$ attempts, π_J projects the pipeline to `NullTree` (**0**), saving both

Stage 1b and all 3 Stage-2 turns. - **Stage 1b (Grafted Chord-Coboundary Materialization, slot arity $A_{1b} = b_1$):** Conditioned on $G_T(x, \hat{p}_T) = 1$ (so $\hat{p}_T = p_T^*$ is verified), the host grafts p_T^* as an immutable `LeafTextSpan` (PromptColour.SubstitutionStep) into the Stage-1b prompt, listing for each chord $g_m = (u_m \to v_m) \in E \setminus T$ its already-verified endpoint potentials $p_T^*(v_m)$ and $p_T^*(u_m)$, and proposes $\hat{c}'_{E\setminus T} = (\hat{c}'_{g_1}, \dots, \hat{c}'_{g_{b_1}}) \in \mathbb{Z}^{b_1}$ under `gbnf_slots = b_1`, checked by the **chord-coboundary guard**

$$G_C(x, p_T^*, \hat{c}'_{E\setminus T}) = 1 \iff \bigwedge_{m=1}^{b_1} \Big(\hat{c}'_{g_m} = p_T^*(v_m) - p_T^*(u_m) \Big).$$

Theorem 8 (Two-Stage Tree-Then-Chord `P*` Slot-Arity Reduction & Kernel Closure). For every connected graph $G = (V, E)$ with spanning tree T ($|T| = |V| - 1$) and $b_1 = |E \setminus T| \geq 1$ chords:

1. **Per-Turn Slot-Arity Reduction:** The maximum integer-slot arity across Stage 1a and Stage 1b satisfies
   $$A_{\text{2-stage}}(G) := \max(|V| - 1,\ b_1) \leq (|V| - 1) + 2b_1 - \min(|V| - 1, b_1) - b_1 < A_{\text{1-stage}}(G),$$
   yielding strict arity reductions of $2b_1$ slots in Stage 1a and $|V| - 1 + b_1$ slots in Stage 1b (specifically $(13, 17, 16, 21, 21) \to (5, 7, 6, 9, 7)$ across $K_{3,3}, Q_3, K_5, \text{Petersen}, W_7$).
2. **Depth-1 Chord Factorization Conditioned on $G_T = 1$:** Once Stage 1a passes $G_T(x, \hat{p}_T) = 1$, $\hat{p}_T = p_T^* = (d_0|_T)^{-1}(c|_T)$ is uniquely fixed in the Stage-1b prompt context. Consequently, each Stage-1b slot $\hat{c}'_{g_m}$ depends only on a single depth-1 two-integer subtraction $p_T^*(v_m) - p_T^*(u_m)$ of literals already present in the Stage-1b prompt, with zero multi-hop path accumulation and zero cross-chord coupling.
3. **Two-Stage False-Green Kernel Closure:** The joint two-stage guard satisfies
   $$G_T(x, \hat{p}_T) = 1 \ \wedge\ G_C(x, \hat{p}_T, \hat{c}'_{E\setminus T}) = 1 \iff (\hat{p}_T, \hat{c}'_{E\setminus T}) = (p_T^*, d_0|_{E\setminus T} p_T^*),$$
   so the two-stage false-green kernel is empty ($\mathcal{K}_{\text{false}}^{\text{2-stage}}(x) = \emptyset$), and every pair admitted to Stage 2 defines an exact zero-defect coboundary $c' = c|_T \oplus \hat{c}'_{E\setminus T} = d_0 p_T^*$ with $\Phi_{1,w}(c') = 0$ and $\Gamma_2(c') = 0$.

Proof. (1) Since $|V| - 1 \geq 1$ and $b_1 \geq 1$, both $|V| - 1 \leq (|V| - 1) + 2b_1 - 2$ and $b_1 \leq (|V| - 1) + 2b_1 - 2$, so $\max(|V| - 1, b_1) < (|V| - 1) + 2b_1$. (2)–(3) Because T is a spanning tree rooted at v_0 with $p(v_0) = 0$, the restriction $d_0|_T : C^0_{\text{root}=0}(G; \mathbb{Z}) \to C^1(T; \mathbb{Z})$ is a $\mathbb{Z}$-module isomorphism; hence $G_T(x, \hat{p}_T) = 1 \iff d_0|_T \hat{p}_T = c|_T \iff \hat{p}_T = p_T^*$. Conditioned on $\hat{p}_T = p_T^*$, $G_C(x, p_T^*, \hat{c}'_{E\setminus T}) = 1$ holds if and only if $\hat{c}'_{g_m} = p_T^*(v_m) - p_T^*(u_m) = (d_0 p_T^*)_{g_m}$ for each $m \in \{1, \dots, b_1\}$ independently. Combining $c'|_T = c|_T = d_0|_T p_T^*$ and $c'|_{E\setminus T} = d_0|_{E\setminus T} p_T^*$ gives $c' = d_0 p_T^*$, so $\Phi_{1,w}(c') = 0$ and $\Gamma_2(c') = 0$. □

5.7 Theorem 9: Dichotomic Fredholm-alternative `P*` guarding and elimination of the lazy false-refusal attractor (`C005`)

Consider the dichotomic relational consistency task (`C005`): given a finite connected directed graph $G = (V, E)$ with integer edge differences $d \in \mathbb{Z}^E$, a participant must decide whether the system $n_v - n_u = d_{(u,v)}$ ($(u, v) \in E$) is `CONSISTENT` (returning an exact vertex potential $n \in \mathbb{Z}^V$ with $d_0 n = d$) or `INCONSISTENT` (returning a simple directed cycle γ whose signed edge sum $\langle \gamma, d \rangle = \sum_{e \in \gamma} \sigma_{\gamma,e} d_e \neq 0$).

In un-staged one-shot generation, small and medium language models exhibit a systematic **lazy false-refusal (`INCONSISTENT`) attractor**: whenever a model makes a single sign or addition mistake while integrating potentials along G, it encounters a chord $g_m = (u_m \to v_m)$ where its erroneous

potentials fail $\hat{n}_{v_m} - \hat{n}_{u_m} = d_{g_m}$, and therefore declares `INCONSISTENT` on consistent boards (`0/4 = 0.0%` accuracy on consistent boards in both Panel 1 and un-staged Panel 2; see Subsection 6.6).

To eliminate this failure mode, fix a rooted spanning tree $T \subset E$ ($|T| = |V| - 1$) and $b_1 = |E \setminus T|$ chords $g_1, \dots, g_{b_1}$, and require Stage 1 to materialize the $(|V| - 1 + b_1)$-slot tuple $z_1 = (\hat{p}_T, \hat{r}_{E\setminus T}) \in \mathbb{Z}^{|V|-1} \times \mathbb{Z}^{b_1}$ subject to the **Dichotomic Fredholm-Alternative Guard**:

$$G_{\text{Fredholm}}(x, \hat{p}_T, \hat{r}_{E\setminus T}) = 1 \iff \bigwedge_{e=(u,v)\in T} (\hat{p}_T(v) - \hat{p}_T(u) = d_e) \ \wedge \ \bigwedge_{m=1}^{b_1} \Big(\hat{r}_m = d_{g_m} - (\hat{p}_T(v_m) - \hat{p}_T(u_m))\Big).$$

Theorem 9 (Dichotomic Fredholm-Alternative `P*` Guard & False-Verdict Elimination). For every connected directed graph $G = (V, E)$ with spanning tree T, chords $g_1, \dots, g_{b_1} \in E \setminus T$, and edge differences $d \in \mathbb{Z}^E$:

1. **Discrete Fredholm Alternative on $(T, E \setminus T)$:** Let $p_T^* = (d_0|_T)^{-1}(d|_T) \in \mathbb{Z}^V$ ($p_T^*(v_0) = 0$) and let $r_m^* := \langle \gamma_m, d \rangle = d_{g_m} - (p_T^*(v_m) - p_T^*(u_m))$ be the holonomy of the m-th chord-induced cycle γ_m. Then:
$$d \in \operatorname{im}(d_0) \iff r_1^* = \cdots = r_{b_1}^* = 0 \iff d = d_0 p_T^*,$$
and conversely $d \notin \operatorname{im}(d_0)$ if and only if there exists $m^* \in \{1, \dots, b_1\}$ with $r_{m^*}^* \neq 0$, in which case γ_{m^*} is an explicit simple-cycle contradiction witness with $\langle \gamma_{m^*}, d \rangle = r_{m^*}^* \neq 0$.
2. **Elimination of the Lazy False-Refusal Attractor ($\mathcal{K}^{\mathbf{C005}}_{\mathbf{false\text{-}verdict}}(x) = \emptyset$):** For every proposal $z_1 = (\hat{p}_T, \hat{r}_{E\setminus T}) \in \mathbb{Z}^{|V|-1+b_1}$,
$$G_{\text{Fredholm}}(x, \hat{p}_T, \hat{r}_{E\setminus T}) = 1 \iff (\hat{p}_T, \hat{r}_{E\setminus T}) = (p_T^*, r_{E\setminus T}^*).$$
Consequently, under the deterministic Stage-2 branch map
$$\mathcal{B}(\hat{p}_T, \hat{r}_{E\setminus T}) := \begin{cases} (\mathsf{CONSISTENT},\ \hat{p}_T) & \text{if } \hat{r}_{E\setminus T} = 0, \\ (\mathsf{INCONSISTENT},\ \gamma_{m^*},\ \hat{r}_{m^*}) & \text{if } m^* = \min\{m : \hat{r}_m \neq 0\}, \end{cases}$$
no Stage-1 proposal admitted by $G_{\text{Fredholm}} = 1$ can emit a false `INCONSISTENT` verdict on a consistent board or a false `CONSISTENT` verdict on an inconsistent board: the admitted false-verdict set is empty ($\mathcal{K}^{\text{C005}}_{\text{false-verdict}}(x) = \emptyset$).

Proof. (1) Because $\{\gamma_1, \dots, \gamma_{b_1}\}$ is a $\mathbb{Z}$-basis for $H_1(G; \mathbb{Z}) = \ker(\partial_1)$, $d \in \operatorname{im}(d_0) = \ker(d_1)$ holds if and only if $\langle \gamma_m, d \rangle = 0$ for every $m = 1, \dots, b_1$. Along $\gamma_m = g_m - \pi_T(u_m \to v_m)$, telescoping $d|_T = d_0|_T p_T^*$ along the unique tree path $\pi_T(u_m \to v_m)$ gives $\sum_{e \in \pi_T(u_m \to v_m)} \sigma_e d_e = p_T^*(v_m) - p_T^*(u_m)$, so $\langle \gamma_m, d \rangle = d_{g_m} - (p_T^*(v_m) - p_T^*(u_m)) = r_m^*$. Thus $r_{E\setminus T}^* = 0 \iff d = d_0 p_T^*$, and if $r_{m^*}^* \neq 0$ then γ_{m^*} is a simple cycle in $T \cup \{g_{m^*}\}$ with non-zero signed sum $r_{m^*}^*$. (2) Under $G_{\text{Fredholm}} = 1$, the first conjunction forces $\hat{p}_T = p_T^*$ by tree-path induction from v_0, and substituting $\hat{p}_T = p_T^*$ into the second conjunction forces $\hat{r}_m = d_{g_m} - (p_T^*(v_m) - p_T^*(u_m)) = r_m^*$ for every $m = 1, \dots, b_1$. Hence $\hat{r}_{E\setminus T} = 0 \iff r_{E\setminus T}^* = 0 \iff d \in \operatorname{im}(d_0)$, so $\mathcal{B}(\hat{p}_T, \hat{r}_{E\setminus T})$ agrees with the exact cohomological truth on every admitted instance. □

5.8 Theorem 10: Cohomological syndrome decoding $\mathbf{r}^* = S\mathbf{c}$ across Verdict, Single-Edge Repair, and Audit (`C001`)

Consider the distractor-contaminated rank-difference witness game (`C001`): a natural-language prompt contains bracketed distractor notes (`[Note k: ...]`, which belong to the congruence ideal $\mathcal{J}_{\text{distractor}}$) mixed with directed integer rank-difference claims $c_{u,v} \in \mathbb{Z}$ (`B - A = c_{A,B}`, $|V| \in \{8, \dots, 11\}, |E| \in \{12, \dots, 16\}, b_1 \in \{5, 7\}, c_{u,v} \in [-5000, 5000]$) and one of three task obligations: - `verdict`: emit `CONSISTENT` with an exact potential $p : V \to \mathbb{Z}$, or CONTRADICTION with an explicit simple cycle whose signed edge sum is non-zero; - `repair`: identify a single corrupted

edge $e^* \in E$, its replacement value $c'_{e^*} \in \mathbb{Z}$, and the resulting exact potential $p' : V \to \mathbb{Z}$ on $(E \setminus \{e^*\}) \cup \{(e^*, c'_{e^*})\}$; - `audit`: inspect a proposed submission (which may contain adversarial off-by-one cycle claims, flipped edge orientations, or deceptive CONTRADICTION claims on consistent graphs) and emit `ACCEPT` or `REJECT` with a verified correction witness.

Fix a rooted spanning tree $T \subset E$ ($|T| = |V|-1$) and chord set $E \setminus T = \{e_1, \dots, e_{b_1}\}$ ($e_m = u_m \to v_m$). For each chord $e_m \in E \setminus T$ and each tree edge $e \in T$, define the signed tree-path incidence

$$S_{m,e} := \begin{cases} +1 & \text{if } e \text{ is traversed forward along the unique path } \pi_T(u_m \to v_m), \\ -1 & \text{if } e \text{ is traversed backward along } \pi_T(u_m \to v_m), \\ 0 & \text{if } e \notin \pi_T(u_m \to v_m), \end{cases}$$

and form the **chord-induced cycle-edge incidence matrix**

$$S := [\, S_T \mid -I_{b_1} \,] \in \{-1, 0, +1\}^{b_1 \times |E|},$$

whose m-th row $S_{m,\cdot}$ is the signed indicator of the chord-induced simple cycle $\mathcal{C}_T(e_m) = \pi_T(u_m \to v_m) \cup \{v_m \to u_m\}$.

Theorem 10 (Cohomological Syndrome Decoding Across Verdict, Single-Edge Repair, and Audit). Suppose Stage 1a materializes $\hat{p}_T \in \mathbb{Z}^{|V|-1}$ passing $G_T(\mathbf{c}, \hat{p}_T) = 1$ (so $\hat{p}_T = p^*_T = B_T^{-1}\mathbf{c}|_T$), and Stage 1b materializes $\hat{\mathbf{c}}_{E\setminus T} \in \mathbb{Z}^{b_1}$ passing $G_C(p^*_T, \hat{\mathbf{c}}_{E\setminus T}) = 1$ (so $\hat{c}_m = \hat{c}^*_m := p^*_T(v_m) - p^*_T(u_m)$ for $m = 1, \dots, b_1$). Define the **cohomological syndrome vector**

$$\mathbf{r}^* := \hat{\mathbf{c}}^* - \mathbf{c}|_{E\setminus T} \in \mathbb{Z}^{b_1}.$$

Then:

1. **Exact Cycle-Space Parity Check:** $\mathbf{r}^* = S\mathbf{c}$ identically, and $\ker(S) = \mathrm{im}(d_0) = \{\mathbf{c} \in \mathbb{Z}^E : \Phi_1(\mathbf{c}) = 0\}$. Consequently, for `verdict` tasks, $\mathbf{r}^* = \mathbf{0} \iff \Phi_1(\mathbf{c}) = 0$ (`CONSISTENT` with exact potential p^*_T), and $\mathbf{r}^* \neq \mathbf{0} \iff \Phi_1(\mathbf{c}) > 0$ (CONTRADICTION, where any m^* with $r^*_{m^*} \neq 0$ yields the certified simple cycle $\mathcal{C}_T(e_{m^*})$ with signed discrepancy $r^*_{m^*} \neq 0$).
2. **Single-Edge Syndrome Decoding (Chord & Spanning-Tree Perturbations):** Let $\mathbf{c} = \mathbf{c}^{\text{true}} + \delta\mathbf{e}_{e^*}$ be obtained from a consistent cochain $\mathbf{c}^{\text{true}} = d_0 p^{\text{true}} \in \mathrm{im}(d_0)$ by corrupting a single edge $e^* \in E$ by $\delta = c_{e^*} - c^{\text{true}}_{e^*} \neq 0$. Then a candidate single-edge replacement (e, c'_e) with shift $\hat{\delta} := c_e - c'_e \in \mathbb{Z} \setminus \{0\}$ yields a globally consistent repaired cochain $\mathbf{c}' = \mathbf{c} - \hat{\delta}\mathbf{e}_e \in \mathrm{im}(d_0)$ if and only if the syndrome $\mathbf{r}^*$ is the $\hat{\delta}$-scaled e-th column of S:
$$\mathbf{r}^* = \hat{\delta}\, S_{\cdot,e}.$$
In particular:
   - **Case 2a (Chord Perturbation, $e^* = e_m \in E \setminus T$):** Since $S_{\cdot,e_m} = -\mathbf{e}_m \in \mathbb{Z}^{b_1}$, a chord corruption has unit Hamming support $\mathrm{supp}(\mathbf{r}^*) = \{m\}$ with $r^*_m = -\delta$, and is repaired by $c'_{e_m} = c_{e_m} + r^*_m = \hat{c}^*_m$ with unchanged potential $p' = p^*_T$.
   - **Case 2b (Spanning-Tree Perturbation, $e^* \in T$):** A tree-edge corruption propagates across the tree-cut incidence set $\mathrm{supp}(\mathbf{r}^*) = \{m \in \{1, \dots, b_1\} : e^* \in \pi_T(u_m \to v_m)\}$ with $r^*_m = \delta S_{m,e^*} \in \{\pm\delta\}$, and is decoded by matching $\mathbf{r}^* = \hat{\delta} S_{\cdot,e^*}$ ($c'_{e^*} = c_{e^*} - \hat{\delta}$) and shifting the subtree potential $p' = B_T^{-1}(\mathbf{c}|_T - \hat{\delta}\mathbf{e}_{e^*})$.
3. **Adversarial Audit Completeness:** For any `audit` submission proposing a potential, contradiction cycle, or single-edge repair, re-evaluating the proposal against $(p^*_T, \mathbf{r}^* = S\mathbf{c})$ rejects all off-by-one cycle claims, flipped-sign edge traversals, and deceptive CONTRADICTION claims on $\mathbf{r}^* = \mathbf{0}$ graphs.

Proof. (1) Along each chord $e_m = (u_m \to v_m)$, telescoping $d_0 p^*_T = \mathbf{c}|_T$ along $\pi_T(u_m \to v_m)$ gives

$\widehat{c}^*_m = p^*_T(v_m) - p^*_T(u_m) = \sum_{e\in T} S_{m,e}c_e = (S_T\mathbf{c}|_T)_m$. Subtracting c_{e_m} gives $r^*_m = (S_T\mathbf{c}|_T)_m - c_{e_m} = (S\mathbf{c})_m$. Since the rows of S form a basis of the cycle space $Z_1(G;\mathbb{Z})$, $S\mathbf{c} = \mathbf{0} \iff \mathbf{c} \in \mathrm{im}(d_0)$. (2) By linearity of S, for $\mathbf{c}' = \mathbf{c} - \hat{\delta}\mathbf{e}_e$, we have $S\mathbf{c}' = S\mathbf{c} - \hat{\delta}S_{\cdot,e} = \mathbf{r}^* - \hat{\delta}S_{\cdot,e}$. Thus $\mathbf{c}' \in \mathrm{im}(d_0) = \ker(S) \iff \mathbf{r}^* = \hat{\delta}S_{\cdot,e}$. When $e = e_m \in E \setminus T$, column $S_{\cdot,e_m} = -\mathbf{e}_m$, so $\mathbf{r}^* = -\hat{\delta}\mathbf{e}_m$ and $c'_{e_m} = c_{e_m} - \hat{\delta} = c_{e_m} + r^*_m = \widehat{c}^*_m$. When $e \in T$, column $S_{\cdot,e}$ has non-zero entries ± 1 on every chord cycle whose tree path traverses e. (3) follows immediately from (1) and (2). □

5.9 Theorem 11: Tree-coboundary defect migration vs. path-integrated B_T^{-1} guard diagnostics

Why does Stage 1a require a **path-integrated** diagnostic guard rather than an **edge-local** coboundary diagnostic on multi-hop spanning trees with 4-digit claims? Order the non-root vertices $V \setminus \{v_0\} = \{v_1, \ldots, v_{|V|-1}\}$ in BFS topological order along the rooted spanning tree T, and let $B_T := d_0|_T \in \{-1, 0, +1\}^{(|V|-1)\times(|V|-1)}$ be the rooted tree coboundary matrix. Its inverse $B_T^{-1} \in \{0, +1\}^{(|V|-1)\times(|V|-1)}$ is the lower-triangular **root-to-vertex path-incidence matrix**:

$$(B_T^{-1})_{w,e} = \mathbb{1}\big[e \in \pi_T(v_0 \to w)\big], \qquad p^*_T(w) = (B_T^{-1}\mathbf{c}|_T)_w = \sum_{e\in\pi_T(v_0\to w)} c_e.$$

For any Attempt-1 proposal $\hat{p}^{(1)}_T \in \mathbb{Z}^{|V|-1}$, define its **edge-local coboundary residual** $\mathbf{e}^{(1)}_T \in \mathbb{Z}^{|V|-1}$ and its **path-integrated potential residual** $\delta^{(1)}_T \in \mathbb{Z}^{|V|-1}$ by

$$\mathbf{e}^{(1)}_T := B_T\hat{p}^{(1)}_T - \mathbf{c}|_T, \qquad \delta^{(1)}_T := \hat{p}^{(1)}_T - B_T^{-1}\mathbf{c}|_T = B_T^{-1}\mathbf{e}^{(1)}_T.$$

Theorem 11 (Tree-Coboundary Defect Migration Under Edge-Local Retry vs. One-Step Convergence Under Path-Integrated B_T^{-1} Diagnostics). Suppose Attempt 1 commits a single arithmetic integration error $\Delta_{v^*} \neq 0$ at an internal tree edge $e^* = (\mathrm{parent}(v^*) \to v^*) \in T$ while integrating all remaining tree edges accurately: $(\mathbf{e}^{(1)}_T)_{e^*} = \Delta_{v^*} \neq 0$ and $(\mathbf{e}^{(1)}_T)_e = 0$ for all $e \in T \setminus \{e^*\}$. Let $\mathrm{Subtree}_T(v^*) := \{w \in V \setminus \{v_0\} : e^* \in \pi_T(v_0 \to w)\}$ be the descendant subtree rooted at v^*, and suppose $|\,\mathrm{Subtree}_T(v^*)| \geq 2$ (so v^* has at least one child in T). Then:

1. **Subtree Shift vs. Single-Edge Local Syndrome:** Although the potential error $\delta^{(1)}_T(w) = (B_T^{-1}\mathbf{e}^{(1)}_T)_w = \Delta_{v^*} \neq 0$ is non-zero at **every** descendant $w \in \mathrm{Subtree}_T(v^*)$, the edge-local coboundary residual $\mathbf{e}^{(1)}_T$ is non-zero **only** at e^*: for every strict descendant edge $e = (u \to w)$ inside $\mathrm{Subtree}_T(v^*)$, both endpoints are shifted by $+\Delta_{v^*}$, so the local difference cancels:
$$\hat{p}^{(1)}_T(w) - \hat{p}^{(1)}_T(u) = (p^*_T(w) + \Delta_{v^*}) - (p^*_T(u) + \Delta_{v^*}) = c_e \implies (\mathbf{e}^{(1)}_T)_e = 0.$$
2. **Coboundary Defect Migration Under Coordinate-Local Retry:** If a retry diagnostic reports only the coordinates where $\mathbf{e}^{(1)}_T \neq 0$ (namely slot v^*) and the model updates only the flagged slot $\hat{p}^{(2)}_T(v^*) = p^*_T(v^*)$ while retaining its unflagged slots $\hat{p}^{(2)}_T(w) = \hat{p}^{(1)}_T(w) = p^*_T(w) + \Delta_{v^*}$ for $w \in \mathrm{Subtree}_T(v^*) \setminus \{v^*\}$, then Retry 1 **fails** G_T: the coboundary defect migrates from e^* to every child edge $e_{\mathrm{child}} = (v^* \to w)$ of v^*:
$$(\mathbf{e}^{(2)}_T)_{e^*} = 0, \qquad (\mathbf{e}^{(2)}_T)_{(v^*\to w)} = (p^*_T(w) + \Delta_{v^*}) - p^*_T(v^*) - c_{(v^*\to w)} = \Delta_{v^*} \neq 0.$$
More generally, coordinate-local repair of $\mathbf{e}_T \neq 0$ requires at least $\mathrm{depth}(\mathrm{Subtree}_T(v^*)) + 1$ sequential rounds to push the coboundary defect off the leaves of T.
3. **One-Retry Convergence Under Path-Integrated B_T^{-1} Diagnostics:** Because $\mathrm{supp}(\delta^{(1)}_T) = \mathrm{Subtree}_T(v^*)$, the path-integrated guard diagnostic $\hat{p}^{(1)}_T(w) \neq (B_T^{-1}\mathbf{c}|_T)_w$ flags v^* and all of its affected descendants $w \in \mathrm{Subtree}_T(v^*)$ simultaneously in a single diagnostic turn, eliminating subtree defect migration.

Proof. (1) Since $\mathbf{e}_T^{(1)} = \Delta_{v^*}\mathbf{e}_{e^*}$, multiplying by B_T^{-1} gives $\delta_T^{(1)}(w) = (B_T^{-1})_{w,e^*}\Delta_{v^*} = \Delta_{v^*}\mathbb{1}[w \in \mathrm{Subtree}_T(v^*)]$. Applying B_T recovers $\mathbf{e}_T^{(1)} = B_T\delta_T^{(1)} = \Delta_{v^*}\mathbf{e}_{e^*}$, which vanishes on every edge $e \neq e^*$. (2) Updating only coordinate v^* sets $\delta_T^{(2)}(v^*) = 0$ while leaving $\delta_T^{(2)}(w) = \Delta_{v^*}$ for $w \in \mathrm{Subtree}_T(v^*) \setminus \{v^*\}$. Applying B_T to $\delta_T^{(2)}$ gives $(\mathbf{e}_T^{(2)})_{(u\to w)} = \delta_T^{(2)}(w) - \delta_T^{(2)}(u)$, which equals $\Delta_{v^*} - 0 = \Delta_{v^*} \neq 0$ for every child edge $(v^* \to w)$ with $w \in \mathrm{Subtree}_T(v^*)$. By induction on subtree depth, each coordinate-local retry advances the zero-error wavefront by one tree level, requiring $\mathrm{depth}(\mathrm{Subtree}_T(v^*)) + 1$ rounds. (3) Directly from $\mathrm{supp}(\delta_T^{(1)}) = \{w : \hat{p}_T^{(1)}(w) \neq p_T^*(w)\} = \mathrm{Subtree}_T(v^*)$. □

5.10 Theorem 12: Independent exact cohomology certificate verification across C011–C020 and elimination of status-label false-SOLVED admissions

Consider the ten-challenge cohomology suite `b_cohomology` (C011–C020: ℓ_1 cycle-rank coboundary repair C011, cellular sheaf cohomology C012, simplicial Smith Normal Form torsion C013, weighted 2-complex discrete Hodge decomposition C014, $\mathbb{F}_2$ persistent homology $R = D \cdot V$ C015, Alexander–Whitney cup-product ring C016, non-abelian lattice gauge holonomy C017, signed-graph switching frustration C018, H^2 cochain extension obstruction C019, and $\mathbb{F}_2$ double-complex spectral sequences C020). For each challenge $m \in \{11, \ldots, 20\}$ and instance x, a candidate JSON certificate $z = (\hat{\sigma}, \hat{w})$ pairs a self-reported status label $\hat{\sigma} \in \{\mathsf{SOLVED}, \mathsf{OUT_OF_SCOPE}, \mathsf{UNKNOWN}\}$ with a structured witness payload $\hat{w}$.

Let $G_{\mathrm{status}}(x, z) := \mathbb{1}[\hat{\sigma} = \mathsf{SOLVED}]$ be the **status-label guard** that trusts the model's self-reported `"status": "SOLVED"` field once JSON schema parsing succeeds, and let $G_{\mathrm{cert}}^{(m)}(x, z) \in \{0, 1\}$ (check_certificate) be the **independent exact certificate guard** that recomputes the defining algebraic identities of challenge m from $(x, \hat{w})$ over $\mathbb{Q}$, $\mathbb{Z}$, $\mathbb{F}_2$, or finite group tables without calling the solver and without trusting $\hat{\sigma}$.

Theorem 12 (Status-Label False-SOLVED Kernel vs. Independent Exact Certificate Closure Across C011–C020). For every cohomology challenge $m \in \{11, \ldots, 20\}$ and valid solvable instance $x \in \mathcal{D}_{\mathrm{solved}}^{(m)}$:

1. **Unbounded Status-Label False-SOLVED Kernel:** The set of schema-valid JSON replies that pass $G_{\mathrm{status}}(x, z) = 1$ while violating the algebraic witness relation $R^{(m)}(x, \hat{w})$,
$$\mathcal{K}_{\text{false-status}}^{(m)}(x) := \{z = (\mathsf{SOLVED}, \hat{w}) : G_{\mathrm{status}}(x, z) = 1 \ \wedge \ \neg R^{(m)}(x, \hat{w})\},$$
is non-empty (and infinite whenever $\hat{w}$ carries integer or rational cochain fields). On accounted 7B open-weight evaluation (`B-COH-S1`, `qwen/qwen-2.5-7b-instruct`), Attempt 1 lands in $\mathcal{K}_{\text{false-status}}^{(m)}(x)$ on **`10/10 (100.0%)`** of challenges C011–C020.
2. **Exact Certificate Guard Closure:** Under the independent verifier $G_{\mathrm{cert}}^{(m)}(x, z) = 1 \iff \hat{\sigma} = \mathsf{SOLVED} \wedge R^{(m)}(x, \hat{w})$ (and for out-of-scope inputs $x \in \mathcal{D}_{\mathrm{oos}}^{(m)}$, $\hat{\sigma} = \mathsf{OUT_OF_SCOPE} \wedge R_{\mathrm{oos}}^{(m)}(x, \hat{w})$), the admitted false-SOLVED kernel is empty:
$$\{z : G_{\mathrm{cert}}^{(m)}(x, z) = 1 \ \wedge \ \neg R^{(m)}(x, \hat{w})\} = \emptyset,$$
and any persistent certificate failure after $R \geq 1$ bounded diagnostic attempts is absorbed by $\pi_{\mathcal{J}}$ into `NullTree` (**0**) with zero false SOLVED admissions to Stage 2.

Proof. (1) Changing any numerical coordinate of a valid witness w^* (e.g., replacing $R = D \cdot V \pmod 2$ by $R = 0$ in C015, or reporting `frustration_index = 0` with an unbalanced switching assignment $\sigma \in \{\pm 1\}^V$ in C018) while keeping `"status": "SOLVED"` preserves $G_{\mathrm{status}}(x, z) = 1$ and violates $R^{(m)}(x, \hat{w})$. (2) By definition of check_certificate, $G_{\mathrm{cert}}^{(m)}(x, z) = 1$ holds only when every defining equation of $R^{(m)}(x, \hat{w})$ holds identically; if $G_{\mathrm{cert}}^{(m)} = 0$ on all R attempts, Theorem 3's ideal

absorption maps the Stage-2 graft to `NullTree`. □

6. Historical mechanism results: causal slot-factorization and guarded retry across six C004 trials, three C006 multi-cycle trials, one C005 dichotomic trial, one C001 syndrome trial, and one `b_cohomology` (`C011`–`C020`) campaign

Current evidence status (2026-10-23): The original mechanism receipt files referenced by the companion are unavailable in this checkout and the original working directory. The values retained in this section and the empirical conclusions below are historical assertions, not revalidated measur ements. They supply no evidence for the new harness campaign. The finite examples can be checked separately with `code/replay.py --finite-only`; that command reports `internal_interventions: NOT_REPLAYED`. Missing historical evidence is not a pass.

6.1 Why single-turn zero-scratchpad operad prompts fail and `P*` cuts help

Consider an autoregressive transformer P_θ tasked with emitting the k-slot repaired cochain $y = (c'_0, \ldots, c'_{k-1}) \in \mathcal{L}_k$ for a directed k-cycle $x = (c, w)$. Under Theorem 2, computing y requires latent intermediate invariants $z = (s, i^*, c_{i^*}, c'_{i^*})$ where $s = \sum_i c_i$, $i^* = \arg\min_i w_i$, $c_{i^*} = c[i^*]$, and $c'_{i^*} = c_{i^*} - s$.

- **Single-Turn Zero-Scratchpad Regime (`operad_grafted_grammar_projection`):** When the entire operad tree is rendered in a single turn under Layer-2 grammar mask $\mathcal{L}_k$ (`gbnf_slots=k`), the first output token $y_0 = c'_0$ must be emitted before any token of z has been written to the causal context:
$$P_{\text{1-turn}}(y \mid x) = P_\theta(y_0 \mid x) \prod_{t=1}^{k-1} P_\theta(y_t \mid x, y_{<t}).$$
Whenever $i^* = 0$, $y_0 = c_0 - s$ already depends on the global cycle sum s and minimum-weight index i^*, forcing the model to compute both inside a single forward pass while decoding an ASCII operad tree container.
- **Multi-Turn `P*` Staged Operad Materialization (`p_star_operad_slot_materiali zation`):** Cutting the operad tree at an intermediate SubstitutionStep operator f_1 (Stage 1, grammar $\mathcal{L}_m$) and grafting the materialized output z as an immutable `LeafTextSpan` into Stage 2 ($f_2 \circ_i z$, grammar $\mathcal{L}_k$) factorizes the conditional generation law in topological dependency order:
$$P_{\text{P*}}(z, y \mid x) = P_\theta(z_0 \mid x) \prod_{j=1}^{m-1} P_\theta(z_j \mid x, z_{<j}) \cdot \mathbb{1}[z \notin \mathcal{J}] \cdot \prod_{t=0}^{k-1} P_\theta(y_t \mid x, f_2 \circ_i z, y_{<t}).$$
Each intermediate slot z_j is conditioned via causal self-attention on the already-emitted prefix $z_{<j}$, and Stage 2 reduces to substituting the grafted value c'_{i^*} at index i^* while copying c_t for $t \neq i^*$.

6.2 Empirical results across five bounded C004 trials

We evaluated this hierarchy across $N = 8$ deterministic C004 directed k-cycle cases ($k \in \{3, 4, 5, 6\}$, two cases per slot count) on two strictly separated panels (never pooled):

- **Panel 1 (Local HF Checkpoint):** Pinned `Qwen/Qwen2.5-0.5B-Instruct` (revision 7ae557604adf67be50417f59c2c2f167def9a775, CPU torch.float32, 0 provider calls, `$0`).
- **Panel 2 (Accounted OpenRouter):** openai/gpt-4o-mini at temperature=0.0 under strict `json_schema` regex slot projection and native Rust verify_decode_receipt attestation.

Table 1: Panel 1 — Pinned local Qwen/Qwen2.5-0.5B-Instruct ($N = 8$ C004 cases)

Trial & Arm	Layer-2 Parseable (parse_integer_slots)	Certified ExactCoboundary ($\Phi_1 = 0, \Gamma_2 = 0$)	Exact ℓ_1-Optimal Repair	Key Mechanism Observation
Trial 1 (c004_paired_benchmark_receipt.json)				
`free` (unconstrained greedy)	`0/8` (`0.0%`)	`0/8` (`0.0%`)	`0/8` (`0.0%`)	Emits prose explanations (`To find the minimum...`).
`f_null` (no-op logit hook)	`0/8` (`0.0%`)	`0/8` (`0.0%`)	`0/8` (`0.0%`)	`8/8` (`100.0%`) token-for-token identical to `free` (Proposition 4.3).
`carrier` (`[0-9\- ]` vocab filter)	`6/8` (`75.0%`)	`0/8` (`0.0%`)	`0/8` (`0.0%`)	Fails slot count on `2/8` cases (`15` slots for $k = 3$; `20` slots for $k = 6$).
`grammar` (`gbnf_slots=k` DFA mask)	`8/8` (`100.0%`)	`0/8` (`0.0%`)	`0/8` (`0.0%`)	`100%` syntax admissibility (Proposition 4.1); `0/8` zero holonomy.
Trial 2 (operad_cochain_ablation_receipt.json)				
operad_grafted_grammar_projection (1-turn)	`8/8` (`100.0%`)	`0/8` (`0.0%`)	`0/8` (`0.0%`)	Zero-scratchpad 1-turn operad tree (`0/8` coboundary).
Trial 3 (p_star_operad_materialization_receipt.json)				
multi_turn_p_star_operad_slot_materialization (3-slot Stage 1 `(s, i*, c'_{i*})`)	`8/8` (`100.0%`)	**`1/8` (`12.5%`)**	**`1/8` (`12.5%`)**	Solved `c004_k3_01` (`-1 -1 2`, $\Phi_1 = 0$, optimal cost `5`).
Trial 4 (p_star_4slot_binding_receipt.json)				

Trial & Arm	Layer-2 Parseable (parse_integer_slots)	Certified ExactCoboundary ($\Phi_1 = 0, \Gamma_2 = 0$)	Exact ℓ_1-Optimal Repair	Key Mechanism Observation
p_star_4slot_index_value_binding (4-slot Stage 1 `(s, i*, c_{i*}, c'_{i*})`) **Trial 5 (p_star_verified_guard_receipt.json, C004-S5)**	`8/8` (`100.0%`)	`0/8` (`0.0%`)	`0/8` (`0.0%`)	`0.5B` model failed cycle-sum arithmetic in slot 0 (`0/8` sum exact).
p_star_verified_stage1_complete_guard (G_{complete} + 1 retry + `NullTree` abstention)	`8/8` Stage 1 (`0/8` Stage 2 admitted)	`0/8` (`0.0%`)	`0/8` (`0.0%`)	**Prevented `8/8` (`100%`) wasted Stage-2 calls** via STAGE1_GUARD_NULL_TREE_ABSTENTION (Theorem 5.4).
no_model_exact_cochain_baseline (`NO_LLM` oracle)	`8/8` (`100.0%`)	`8/8` (`100.0%`)	`8/8` (`100.0%`)	Deterministic host lightest-edge coboundary control (Theorem 2).

Table 2: Panel 2 — Accounted OpenRouter openai/gpt-4o-mini ($N = 8$ C004 cases)

Trial & Arm	Calls / Settled Spend	Parseable	Stage-1 Invariant Accuracy	Certified ExactCoboundary ($\Phi_1 = 0, \Gamma_2 = 0$)	Exact ℓ_1-Optimal Repair
Trial 1 (c004_paired_benchmark_receipt.json)					
hosted_unconstrained	8 calls / $0.0011	0/8 (0.0%)	N/A	0/8 (0.0%)	0/8 (0.0%)
hosted_slot_constrained (gbnf_slots=k)	8 calls / $0.0012	8/8 (100.0%)	N/A	2/8 (25.0%)	0/8 (0.0%)
Trial 2 (operad_cochain_ablation_receipt.json)					
plain_grammar_projection (1-turn)	8 calls / $0.0011	8/8 (100.0%)	N/A	5/8 (62.5%)	1/8 (12.5%)
operad_grafted_grammar_projection (1-turn, 0 intermediate slots)	8 calls / $0.0018	8/8 (100.0%)	N/A (0 intermediate slots)	0/8 (0.0%)	0/8 (0.0%)
Trial 3 (p_star_operad_materialization_receipt.json)					
plain_grammar_projection (1-turn matched control)	8 calls / $0.0011	8/8 (100.0%)	N/A	6/8 (75.0%)	1/8 (12.5%)
multi_turn_p_star_operad_slot_materialization (2-turn P*, 3-slot (s, i*, c'_{i*}))	16 calls / $0.0031	8/8 (100.0%)	s: 7/8 (87.5%), i*: 5/8 (62.5%), c'_{i*}: 2/8 (25.0%), all 3: 2/8 (25.0%)	3/8 (37.5%)	**2/8 (25.0%)**
Trial 4 (p_star_4slot_binding_receipt.json)					
p_star_4slot_index_value_binding (2-turn P*, 4-slot (s, i*, c_{i*}, c'_{i*}), unverified)	16 calls / $0.002994	8/8 (100.0%)	s: 7/8 (87.5%), i*: 5/8 (62.5%), c_{i*}: 5/8 (62.5%), c'_{i*}: 4/8 (50.0%), all 4: 3/8 (37.5%)	5/8 (62.5%)	**3/8 (37.5%)**

Trial & Arm	Calls / Settled Spend	Parseable	Stage-1 Invariant Accuracy	Certified ExactCoboundary ($\Phi_1 = 0, \Gamma_2 = 0$)	Exact ℓ_1-Optimal Repair
Trial 5 (p_star_verified_ guard_receipt.json, C004-S5)					
p_star_verified_st age1_complete_gu ard (G_{complete} + 1 diagnostic retry, $R = 2$**)**	7 incremental calls / $0.001472	5/5 admitted (3 abstained via NullTree)	s: **7/8 (87.5%)**, i*: **7/8 (87.5%)**, c_{i*}: **7/8 (87.5%)**, c'_{i*}: **5/8 (62.5%)**, all 4: **5/8 (62.5%)**	**5/8 (62.5%)** (5/5 = **100% of admitted**)	**5/8 (62.5%)** (5/5 = **100% conditional precision**)

6.3 Diagnostic analysis of the 5-trial C004 progression

1. **Layer 2 (`gbnf_slots=k`) is necessary for syntax admissibility but insufficient for mathematical validity:** Across both local Qwen2.5-0.5B-Instruct and hosted openai/gpt-4o-mini, unconstrained decoding (`free` / hosted_unconstrained) achieved `0/8` (`0.0%`) parseable integer-slot tuples, whereas Layer-2 grammar projection achieved `8/8` (`100.0%`) parseable tuples. However, grammar compliance alone yielded `0/8` ℓ_1-optimal repairs in Trial 1, confirming Theorem 1's separation between surface decoding (D) and Layer-3 mathematical checking (C).
2. **Single-turn zero-scratchpad operad trees induce a causal ordering failure:** In Trial 2, rendering a 6-operator ColouredOperadPrompt tree inside a single zero-scratchpad turn (operad_grafted_grammar_projection) collapsed openai/gpt-4o-mini from `5/8` ExactCoboundary (`1/8` optimal) down to `0/8` ExactCoboundary (`0/8` optimal), with `4/8` cases copying the unrepaired input claims verbatim (`weighted_l1 = 0`).
3. **2-turn `P*` staged operad slot materialization (`(s, i*, c'_{i*})`) restores and doubles ℓ_1-optimal repair:** In Trial 3, cutting the operad tree so Stage 1 materializes `(s, i*, c'_{i*})` (`gbnf_slots=3`) before host slot-grafting ($\circ_i$) into Stage 2 (`gbnf_slots=k`) raised openai/gpt-4o-mini from `0/8 -> 3/8` ExactCoboundary and `0/8 -> 2/8` (`25.0%`) ℓ_1-optimal (`c004_k3_02`, `c004_k4_02`), and raised local Qwen2.5-0.5B-Instruct from `0/8 -> 1/8` (`12.5%`) optimal (`c004_k3_01`). Crucially, Stage-1 grading exposed the exact remaining bottleneck: openai/gpt-4o-mini computed s accurately on `7/8` (`87.5%`) cases and i^* accurately on `5/8` (`62.5%`) cases, but failed on slot 2 ($c'_{i^*} = c_{i^*} - s$) on `3/5` of the exact-`(s, i*)` cases because it had to dereference c_{i^*} at the just-emitted index i^* and subtract s in a single token step.
4. **4-slot Stage-1 index-value binding (`(s, i*, c_{i*}, c'_{i*})`) resolves the dereference bottleneck and raises ℓ_1-optimality to `3/8` (`37.5%`):** In Trial 4, inserting an explicit slot for the looked-up claim value c_{i^*} (`gbnf_slots=4`) yielded `5/5` (`100%`) exact lookup of c_{i^*} whenever i^* was exact (and `8/8` / `100%` consistent dereferencing $c[\widehat{i^*}]$ at the model's chosen index $\widehat{i^*}$). This doubled Stage-1 c'_{i^*} accuracy from `2/8` (`25.0%`) to `4/8` (`50.0%`), raised Stage-2 ExactCoboundary certification from `3/8` (`37.5%`) to `5/8` (`62.5%`), and raised exact ℓ_1-optimal repair from `2/8` (`25.0%`) to `3/8` (`37.5%`) (`c004_k4_01`, `c004_k4_02`, `c004_k5_01`), with `3/3` (`100%`) Stage-2 ℓ_1-optimality whenever Stage 1 materialized all 4 slots accurately.
5. **Evaluator-verified Stage-1 guard filtering + 1 diagnostic retry (`C004-S5`) raises exact ℓ_1-optimality to `5/8` (`62.5%`) with `5/5` (`100%`) conditional precision and witnesses Theorem 6:** In Trial 5 (p_star_verified_guard_receipt.json), applying the deterministic complete Stage-1 guard G_{complete} with $R = 2$ (at most 1 diagnostic retry on the 5 cases that failed Attempt 1):
   - Repaired `c004_k3_01` from Attempt 1 `5 1 -1 -6` (GUARD_FAIL_MIN_WEIGHT_INDEX) to Retry 1 `5 0 4 -1`, whose admitted Stage-2 graft emitted `-1 -1 2` (certified ExactCoboundary and ℓ_1-optimal at cost `5`).
   - Repaired `c004_k3_02` from Attempt 1 `4 1 6 -2` (GUARD_FAIL_SUBTRACTION_INVARIANT: $-2+4 = 2 \neq 6$) to Retry 1 `4 1 6 2`, whose admitted Stage-2 graft emitted `-3 2 1` (certified ExactCoboundary and ℓ_1-optimal at cost `4`).
   - Raised Stage-1 all-4-slot accuracy from `3/8` (`37.5%`) to `5/8` (`62.5%`), achieved `5/5` (`100.0%`) **conditional ℓ_1-optimal precision** across all admitted Stage-2 grafts (Theorem 5.2), raised unconditional ℓ_1-optimal repair from `3/8` (`37.5%`) to `5/8` (`62.5%`) (Theorem 5.3), and abstained via `NullTree` on the remaining `3/8` Panel-2 cases and `8/8` Panel-1 cases, preventing `3` and `8` wasted Stage-2 calls respectively (Theorem 5.4).
   - **Concrete Local Green / Global Red Witness for Theorem 6 (`c004_k6_01` retry):**

On `c004_k6_01` ($c = (3, -2, 4, -1, 2, -3)$, $w = (3, 2, 1, 2, 3, 2)$, true sum $s(c) = 3$, $i^* = 2$, $c_2 = 4$), Attempt 1 emitted `-1 1 -2 1` (failing GUARD_FAIL_MIN_WEIGHT_INDEX, GUARD_FAIL_SUBTRACTION_INVARIANT, and GUARD_FAIL_CYCLE_SUM). On Retry 1, openai/gpt-4o-mini emitted `z_1 = (0, 2, 4, 4)`, which **passed the partial guard** $G_{\textbf{partial}}(x, z_1) = 1$ (since $w_2 = 1 = \min_j w_j$, $\hat{c}_2 = 4 = c_2$, and $\hat{c}'_2 + \hat{s} = 4 + 0 = 4 = c_2$), yet belonged to the false-green kernel $\mathcal{K}_{\text{false}}(x)$ at $\Delta s = 0 - 3 = -3 \neq 0$ (Theorem 6.1)! Because G_{complete} checked $\hat{s} = \sum_j c_j = 3$, it caught GUARD_FAIL_CYCLE_SUM and projected Retry 1 to `NullTree` before Stage 2.

6.4 Trial 6 (`C006-S6`): Paired multi-cycle overlap-basis cochain repair on K_4 ($b_1 = 3 \geq 2$)

To test whether `P*` staged slot materialization and complete Stage-1 guard filtering extend from single cycles ($b_1 = 1$) to overlapping multi-cycle complexes ($b_1 = \dim H_1(G) \geq 2$), we executed a bounded paired trial (c006_overlap_basis_p_star_receipt.json) across $N = 4$ deterministic K_4 multi-cycle overlap boards (round5-K4-diagnostic at $\Phi_1 = 6$, C006-v2-r3-s6030 at $\Phi_1 = 16$, C006-v2-r3-s6031 at $\Phi_1 = 9$, and weighted board C006-v2-r3-s6032-weighted at $\Phi_1 = 12$ with $w_T = 3, w_{E \setminus T} = 1$). Each board has $|V| = 4, |E| = 6, b_1 = 3$, and an explicit non-star spanning tree T whose central edge is shared by all 3 chord-induced cycles (`max_tree_edge_cycle_incidence = 3`).

Table 3: Trial 6 (`C006-S6`, $N = 4$ K_4 multi-cycle overlap boards, $b_1 = 3$)

Panel & Arm	Calls / Reserved Spend	Stage-1 $G^{\mathrm{C006}}_{\mathrm{partial}}$ Pass (`Att 1`)	Stage-1 $G^{\mathrm{C006}}_{\mathrm{complete}}$ Pass (`Att 1 -> Retry 1`)	Stage-2 `NullTree` Abstentions / Saved Turns	Solved ACCEPTED_ZERO_DEFECT ($\Phi_1 = 0$)	Conditional Stage-2 Solve Rate
Panel 1 (Qwen/Qwen2.5-0.5B-Instruct, local)						
c006_unstaged_action_generation (3-turn session)	0 provider calls / `$0`	N/A	N/A	`0/4` (0 turns saved)	`0/4` (`0.0%`)	`0/4` (`12/12` turns failed TASK_ENVELOPE)
c006_p_star_staged_guarded_overlap_repair (`P*` + $G^{\mathrm{C006}}_{\mathrm{complete}}$, $R = 2$)	0 provider calls / `$0`	`0/4` (`0.0%`)	`0/4 -> 0/4` (`0.0%`)	**`4/4` (12 Stage-2 turns saved)**	`0/4` (`0.0%`)	N/A (0 corrupted Stage-2 admits)
no_model_exact_c006_baseline (`NO_LLM` oracle)	0 provider calls / `$0`	`4/4` (`100.0%`)	`4/4` (`100.0%`)	`0/4`	**`4/4` (`100.0%`, `4/4` replay verified)**	
Panel 2 (openai/gpt-4o-mini, OpenRouter)						
c006_unstaged_action_generation (3-turn session)	12 calls / `$0.0044`	N/A	N/A	`0/4` (0 turns saved)	**`0/4` (`0.0%`)**	`0/4` (`12/12` rejected: NO_STRICT_DESCENT / AM-4-STATIONARY-REPEAT)
c006_p_star_staged_guarded_overlap_repair (`P*` + $G^{\mathbf{C006}}_{\mathbf{complete}}$, $R = 2$)	14 calls / `$0.0049` (`$0.009252` total reserved)	**`2/4` (`50.0%` false-green witnesses)**	`0/4 -> 2/4` (`50.0%`)	**`2/4` (6 Stage-2 turns saved)**	`2/4` (`50.0%`)	**`2/2` (`100.0%`, `2/2` replay verified)**

Key multi-cycle findings from Trial 6 (`C006-S6`):

1. **Un-staged multi-cycle action generation (`0/4` solved) stalls in stationary repetition:** Without a Stage-1 basis-materialization turn, openai/gpt-4o-mini emitted `{"move": "CUT", "claims": <unchanged initial claims>}` on all 12 turns across all 4 K_4 boards, triggering NO_STRICT_DESCENT and AM-4-STATIONARY-REPEAT at $\Phi_1 \in \{6, 16, 9, 12\}$.
2. **Three empirical multi-cycle Local Green / Global Red witnesses for Theorem 7.2:** On Attempt 1, openai/gpt-4o-mini produced two 9-slot tuples in the partial-guard visibility gap $\mathcal{K}^{\text{C006}}_{\text{false}}(x) \cong \mathbb{Z}^6 \setminus \{0\}$ that satisfied $\hat{c}'_{g_m} + \hat{r}_m = c_{g_m}$ on all 3 chords ($G^{\text{C006}}_{\text{partial}} = 1$) while failing $G^{\text{C006}}_{\text{complete}}$: round5-K4-diagnostic Attempt 1 (`"1 3 5 -2 -2 -4 5 3 6"`) and C006-v2-r3-s6030 Attempt 1 (`"28 -27 16 16 27 12 8 -23 8"`). Moreover, on round5-K4-diagnostic Retry 1, it emitted `"1 3 4 -2 -1 -3 5 2 5"`, where the tree potentials $(1, 3, 4)$ were exact ($\Delta p = (0, 0, 0)$) and $\hat{c}'_{g_m} + \hat{r}_m = c_{g_m}$ held on all 3 chords ($-2+5=3$, $-1+2=1$, $-3+5=2$, so $G^{\text{C006}}_{\text{partial}} = 1$), yet $\Delta r = (-1, 4, -1) \neq 0$ so $\hat{c}'_{g_m} \neq \hat{p}(v_m) - \hat{p}(u_m)$! Checking $G^{\text{C006}}_{\text{complete}}$ caught all 3 false-green tuples before Stage 2.
3. **Triangular chord decoupling + 1 diagnostic retry lifts multi-cycle zero-defect repair from `0/4` to `2/4` (`50.0%`) with `2/2` (`100%`) conditional Stage-2 precision:** On Retry 1, openai/gpt-4o-mini repaired its sign-subtraction errors on both C006-v2-r3-s6031 (`"-4 0 -2 4 2 2 -1 5 1"` → `"-4 0 -2 0 -2 2 3 -5 1"`) and weighted board C006-v2-r3-s6032-weighted (`"2 1 3 1 3 2 3 -1 1"` → `"2 1 3 -1 3 1 5 -5 2"`). Both passed $G^{\text{C006}}_{\text{complete}}$, were grafted via $\circ_i$ into Stage 2 (`ASSERT` → `CUT` → `VERIFY`), and achieved certified ACCEPTED_ZERO_DEFECT ($\Phi_1 = 0$, primal-dual certificate verified, zero-send replay_session verified), while `NullTree` abstention prevented `6` hosted (`12` local) wasted Stage-2 turns.

6.5 Trial 7 (`C006-S7`): Higher-Betti ($b_1 \in \{4, 5, 6, 7\}$) two-stage tree-then-chord `P*` factorization across $K_{3,3}, Q_3, K_5$, Petersen, W_7

To test Theorem 8 across non-planar and higher-Betti multi-cycle topologies, we executed a bounded paired trial (c006_higher_betti_p_star_receipt.json) across $N = 5$ deterministic boards from c006.panels.v2: - C006-v2-r4-s6040 ($K_{3,3}$, $|V| = 6, |E| = 9, b_1 = 4$, initial $\Phi_1 = 14$, single-stage arity 13 → two-stage max arity 5) - C006-v2-r5-s6050 (Q_3, $|V| = 8, |E| = 12, b_1 = 5$, initial $\Phi_1 = 22$, single-stage arity 17 → two-stage max arity 7) - C006-v2-r6-s6060 (K_5, $|V| = 5$, $|E| = 10, b_1 = 6$, initial $\Phi_1 = 18$, single-stage arity 16 → two-stage max arity 6) - C006-v2-r6-s6061 (Petersen, $|V| = 10, |E| = 15, b_1 = 6$, initial $\Phi_1 = 24$, single-stage arity 21 → two-stage max arity 9) - C006-v2-r7-s6070 (W_7, $|V| = 8, |E| = 14, b_1 = 7$, initial $\Phi_1 = 28$, single-stage arity 21 → two-stage max arity 7)

Table 4: Trial 7 (`C006-S7`, $N = 5$ higher-Betti boards, $b_1 \in \{4, 5, 6, 7\}$)

Panel & Arm	Calls / Reserved Spend	Stage-1a G_T Pass (`Att 1 -> Retry 1`)	Stage-1b G_C Pass (`Att 1`)	`NullTree` Abstentions / Saved Turns	Solved ACCEPTED_ZERO_DEFECT ($\Phi_1 = 0$)	Conditional Stage-2 Solve Rate
Panel 1 (Qwen/Qwen2.5-0.5B-Instruct, local)						
c006_unstaged_action_generation (3-turn session)	0 provider calls / `$0`	N/A	N/A	`0/5` (0 turns saved)	`0/5` (`0.0%`)	`0/5` (`0.0%`)
c006_p_star_2stage_tree_chord_guarded_repair (`C006-S7`, $R_{1a} = 2$, $R_{1b} = 2$)	0 provider calls / `$0`	`0/5 -> 0/5` (`0.0%`)	`0/0`	**`5/5` (20 downstream turns saved)**	`0/5` (`0.0%`)	N/A (0 corrupted admits)
no_model_exact_c006_higher_beti_baseline (`NO_LLM` oracle)	0 provider calls / `$0`	`5/5` (`100.0%`)	`5/5` (`100.0%`)	`0/5`	**`5/5` (`100.0%`)**	**`5/5` (`100.0%`, `5/5` replay verified)**
Panel 2 (openai/gpt-4o-mini, OpenRouter)						
c006_unstaged_action_generation (3-turn session)	15 calls	N/A	N/A	`0/5` (0 turns saved)	**`0/5` (`0.0%`, final** $\Phi_1 = (14, 22, 18, 24, 28)$**)**	`0/5` (`0.0%`)
c006_p_star_2stage_tree_chord_guarded_repair (`C006-S7`, $R_{1a} = 2, R_{1b} = 2$)	26 calls (`$0.02294` total reserved $\leq$ $0.050 cap)	**`4/5 -> 5/5` (`100.0%`; Petersen recovered on Retry 1)**	**`5/5` (`100.0%` on Attempt 1,** `0` **retries)**	**`0/5` (`5/5` admitted)**	**`5/5` (`100.0%` across** $K_{3,3}, Q_3, K_5,$ **Petersen**, W_7**)**	**`5/5` (`100.0%`, `5/5` replay verified)**

Key higher-Betti findings from Trial 7 (C006-S7):

1. **Two-stage tree-then-chord factorization lifts higher-Betti solve rate from 0/5 (0.0%) to 5/5 (100.0%) on openai/gpt-4o-mini:** In un-staged action generation, openai/gpt-4o-mini solved 0/5 boards across $b_1 \in \{4, 5, 6, 7\}$, leaving every board at its initial compatibility defect $\Phi_1 \in \{14, 22, 18, 24, 28\}$. Under two-stage P* factorization (Theorem 8.1), Stage 1a (G_T, arity $|V| - 1 \in \{5, 7, 4, 9, 7\}$) passed on Attempt 1 for $K_{3,3}, Q_3, K_5$, and W_7 (4/5 = 80.0%), and on the 10-vertex Petersen cage (C006-v2-r6-s6061, 9 tree-potential slots) a single targeted tree-edge retry (build_c006_stage1a_retry_turn) repaired the failed tree-edge slots to pass G_T (5/5 = 100.0%).
2. **100% Attempt-1 chord-coboundary accuracy once verified tree potentials are grafted (Theorem 8.2):** Once p_T^* was verified by G_T and grafted into Stage 1b, openai/gpt-4o-mini computed all $b_1 \in \{4, 5, 6, 6, 7\}$ repaired chord claims $\widehat{c}'_{g_m} = p_T^*(v_m) - p_T^*(u_m)$ with **5/5 (100.0%) exactness on Attempt 1** (0 Stage-1b retries needed), and all 5/5 (100.0%) Stage-2 3-turn sessions (ASSERT → CUT → VERIFY) terminated in certified ACCEPTED_ZERO_DEFECT ($\Phi_1 = 0$).

6.6 Trial 8 (C005-S1): Dichotomic Fredholm-alternative P* guarded separation across 8 public difference boards

To test Theorem 9 on dichotomic (CONSISTENT vs. INCONSISTENT) verification, we executed a bounded paired trial (c005_dichotomic_fredholm_p_star_receipt.json) across all $N = 8$ public C005 difference boards (pub_cons_01..pub_cons_08, 4 CONSISTENT and 4 INCONSISTENT, $|V| \in \{4, 5, 6\}$, $b_1 \in \{1, 2\}$).

Table 5: Trial 8 (C005-S1, $N = 8$ public C005 difference boards, 4 consistent + 4 inconsistent)

Panel & Arm	Calls / Reserved Spend	Stage-1 G_{Fredholm} Pass (`Att 1 -> Retry 1`)	Consistent Solved (4 boards)	Inconsistent Solved (4 boards)	Total Solved (8 boards)	Lazy False `INCONSISTENT` Count	Admitted False Verdicts	Conditional Stage-2 Precision
Panel 1 (Qwen/Qwen2.5-0.5B-Instruct, local)								
c005_unstaged_consistency_verdict	0 calls / $0	N/A	**0/4 (0.0%)**	4/4 (100.0%)	4/8 (50.0%)	**4/4 (100.0%)**	4/8 (50.0%)	4/8 (50.0%)
c005_p_star_dichotomic_fredholm_guarded (C005-S1, $R = 2$)	0 calls / $0	0/8 -> 0/8 (0.0%)	0/4 (4/4 NullTree)	0/4 (4/4 NullTree)	0/8	**0 (0.0%)**	**0 (0.0%)**	N/A (8/8 NullTree abstentions)
no_model_exact_c005_baseline (NO_LLM oracle)	0 calls / $0	8/8 (100.0%)	4/4 (100.0%)	4/4 (100.0%)	**8/8 (100.0%)**	0	0	**8/8 (100.0%)**
Panel 2 (openai/gpt-4o-mini, OpenRouter)								
c005_unstaged_consistency_verdict	8 calls	N/A	**0/4 (0.0%)**	3/4 (75.0%)	**3/8 (37.5%)**	**3 (75.0% of consistent boards)**	**5/8 (62.5%)**	3/8 (37.5%)
c005_p_star_dichotomic_fredholm_guarded (C005-S1, $R = 2$)	21 calls ($0.003636 total reserved $\leq$ $0.050 cap)	**1/8 -> 6/8 (75.0%; 2/8 NullTree)**	**3/4 (75.0%)**	**3/4 (75.0%)**	**6/8 (75.0%)**	0 (0.0%)	0 (0.0%)	**6/6 (100.0%, 6/6 replay verified)**

Key dichotomic findings from Trial 8 (C005-S1):

1. **Empirical confirmation of the lazy false-refusal attractor in un-staged generation:** In un-staged generation, both Qwen/Qwen2.5-0.5B-Instruct (0/4 consistent, 4/4 lazy false INCONSISTENT) and openai/gpt-4o-mini (0/4 consistent, 3/4 lazy false INCONSISTENT) solved **0.0% of consistent boards**, defaulting to INCONSISTENT whenever their unverified mental tree integration encountered a chord discrepancy.
2. **Elimination of admitted false verdicts and 2x overall solve-rate improvement under G_{Fredholm} (Theorem 9):** Enforcing G_{Fredholm} with at most 1 equation-specific diagnostic retry lifted openai/gpt-4o-mini's Stage-1 pass rate from 1/8 (12.5%) on Attempt 1 to **6/8 (75.0%)** after Retry 1 (3 consistent + 3 inconsistent boards admitted; 2/8 projected to NullTree), raised consistent board accuracy from **0/4 (0.0%) to 3/4 (75.0%)**, doubled overall accuracy from **3/8 (37.5%) to 6/8 (75.0%)** (6/6 = 100.0% conditional precision), and reduced both lazy_false_inconsistent_count and admitted_false_verdict_count to 0 across both panels.

6.7 Trial 9 (C001-S1): Distractor-filtered cochain & cohomological syndrome P* factorization across 12 public and poisoned cases

To test Theorems 10 and 11 on larger graphs with 4-digit integer claims ($c_{u,v} \in [-5000, 5000]$, $|V| \in \{8, 9, 10, 11\}$, $|E| \in \{12, \ldots, 16\}$, $b_1 \in \{5, 7\}$) and bracketed natural-language distractors ([Note k: ...]), we executed a bounded paired trial (c001_distractor_cochain_p_star_receipt.json) across $N = 12$ cases from c001/public_cases.json and c001/poisoned_cases.json: - 4 public verdict cases (c001_verdict_1000..1003: 2 CONSISTENT, 2 CONTRADICTION), - 3 public repair cases (c001_repair_1100..1102: chord perturbations ['H','C'] -> -30 and ['A','G'] -> -3387, plus spanning-tree perturbation ['D','C'] -> 2408 propagating across 4 chord cycles), - 2 public audit cases (c001_audit_1200 ACCEPT, c001_audit_1201 REJECT), and - 3 adversarial poisoned cases (c001_poisoned_almost_valid_4000 with an off-by-one +1 cycle discrepancy, c001_poisoned_flipped_sign_4100 with a flipped edge orientation, and c001_poisoned_deceptive_verdict_4200 with a deceptive CONTRADICTION claim on a consistent graph).

Table 6: Trial 9 (C001-S1, $N = 12$ distractor-filtered 4-digit rank-difference cases, $b_1 \in \{5, 7\}$)

Panel & Arm	Calls / Reserved Spend	Stage-1a G_T Pass (`Att 1 -> Retry 1`)	Stage-1b G_C Pass (`Att 1 -> Retry 1`)	Solved Verdict (6)	Solved Repair (3)	Solved Audit (3)	Solved Public (9) / Poisoned (3)	Total Solved (12)	False `SOLVED` Verdicts
Panel 1 (Qwen/Qwen2.5-0.5B-Instruct, local)									
c001_unstaged_witness_generation (1-shot JSON)	0 calls / $0	N/A	N/A	0/6	0/3	0/3	0/9 / 0/3	0/12 (0.0%)	1/12 (8.3%)
c001_p_star_carrier_cochain_syndrome_guarded (`C001-S1`, $R_{1a} = 2$, $R_{1b} = 2$)	0 calls / $0	0/12 -> 0/12	0/0 (12 saved)	0/6	0/3	0/3	0/9 / 0/3	0/12 (12/12 `NullTree`)	**0/12 (0.0%)**
no_model_exact_c001_baseline (`NO_LLM` oracle)	0 calls / $0	12/12 (100%)	12/12 (100%)	6/6 (100%)	3/3 (100%)	3/3 (100%)	9/9 / 3/3 (100%)	**12/12 (100.0%)**	**0/12 (0.0%)**
Panel 2 (openai/gpt-4o-mini, OpenRouter)									
c001_unstaged_witness_generation (1-shot JSON)	12 calls	N/A	N/A	**0/6 (0.0%)**	**0/3 (0.0%)**	1/3 (33.3%)	1/9 (11.1%) / **0/3 (0.0%)**	**1/12 (8.3%)**	**11/12 (91.7%)**

Panel & Arm	Calls / Reserved Spend	Stage-1a G_T Pass (`Att 1 -> Retry 1`)	Stage-1b G_C Pass (`Att 1 -> Retry 1`)	Solved Verdict (6)	Solved Repair (3)	Solved Audit (3)	Solved Public (9) / Poisoned (3)	Total Solved (12)	False `SOLVED` Verdicts
c001_p_star_carrier_cochain_syndrome_guarded (`C001-S1`, $R_{1a} = 2$, $R_{1b} = 2$)	41 calls (`$0.012427` total reserved $\leq$ $0.050 cap)	`1/12 -> 12/12 (100.0%)`	`6/12 -> 12/12 (100.0%)`	`6/6 (100.0%)`	`3/3 (100.0%)`	`3/3 (100.0%)`	`9/9 (100.0%) / 3/3 (100.0%)`	`12/12 (100.0%)`	`0/12 (0.0%)`

Key findings from Trial 9 (C001-S1):

1. **Collapse of un-staged 1-shot witness generation under 4-digit arithmetic and adversarial distractors (1/12 = 8.3%):** In un-staged generation, openai/gpt-4o-mini solved 0/6 verdict cases, 0/3 repair cases, and 0/3 adversarial poisoned cases (1/12 = 8.3% overall, solving only c001_audit_1200), emitting **11/12 (91.7%) false SOLVED verdicts** whose potentials failed $B\mathbf{p} = \mathbf{c}$ or whose cycles failed simple-cycle non-zero holonomy verification.
2. **Empirical verification of Theorem 11 (Edge-local coboundary defect migration vs. path-integrated B_T^{-1} retry convergence):** Because the spanning-tree claims $c_{u,v} \in [-5000, 5000]$ are 4-digit signed integers across 7 to 10 tree hops, openai/gpt-4o-mini passed Stage 1a (G_T) on Attempt 1 on only 1/12 (8.3%) cases (c001_poisoned_almost_valid_4000). Under an initial edge-local retry diagnostic ($B_T\hat{p}_T^{(1)} \neq \mathbf{c}|_T$), fixing a flagged parent slot v^* while retaining its unflagged descendants $w \in \text{Subtree}_T(v^*) \setminus \{v^*\}$ caused the coboundary defect to migrate to the child edges ($v^* \to w$) (2/12 = 16.7% convergence). Switching to the path-integrated diagnostic $\hat{p}_T^{(1)}(w) \neq (B_T^{-1}\mathbf{c}|_T)_w$ (Theorem 11.3) flagged the entire shifted subtree simultaneously and lifted Stage-1a retry convergence from 1/12 (8.3%) on Attempt 1 (and 2/12 under edge-local retry) to **12/12 (100.0%)** on Retry 1!
3. **Cohomological syndrome decoding $\mathbf{r}^* = S\mathbf{c}$ solves 12/12 (100.0%) across verdict, chord/tree repair, audit, and poisoned traps (Theorem 10):** Stage 1b (G_C) converged from 6/12 (50.0%) on Attempt 1 to **12/12 (100.0%)** after Retry 1. From the verified pair (p_T^*, $\hat{\mathbf{c}}^*$), the syndrome $\mathbf{r}^* = \hat{\mathbf{c}}^* - \mathbf{c}|_{E\setminus T} = S\mathbf{c}$ decoded:
   - all 6/6 verdict cases (including c001_poisoned_almost_valid_4000 and c001_poisoned_deceptive_verdict_4200),
   - all 3/3 single-edge repair cases—both unit-support chord perturbations (c001_repair_1100: ['H', 'C'] -> -30; c001_repair_1102: ['A', 'G'] -> -3387, Theorem 10.2a) and the multi-cycle spanning-tree perturbation (c001_repair_1101: ['D', 'C'] -> 2408 with cutset support of size 4, Theorem 10.2b), and
   - all 3/3 audit cases (including c001_poisoned_flipped_sign_4100), with **0/12 false SOLVED verdicts** across both panels.

6.8 Trial 10 (C004-S6): Held-out 40-case directed cycle ladder ($k \in \{3, \ldots, 12\}$)

To test Theorem 5 and Theorem 6 beyond $k \leq 6$ on a held-out 7B open-weight checkpoint (qwen/qwen-2.5-7b-instruct on OpenRouter, with local hardware VRAM boundary recorded at 4.0 GB RTX 3050 vs. 15.24 GB bf16 requirement alongside local Qwen/Qwen2.5-0.5B-Instruct), we executed the held-out 40-case C004 ladder (c004_40case_hidden_p_star_receipt.json, 4 cases per cycle length $k \in \{3, \ldots, 12\}$):

- **NO_LLM exact baseline (no_model_exact_c004_40case_baseline):** 40/40 (100.0%) certified ExactCoboundary and ℓ_1-optimal (16/16 in $k \in \{3..6\}$, 16/16 in $k \in \{7..10\}$, 8/8 in $k \in \{11, 12\}$).
- **Panel 1 (Qwen/Qwen2.5-0.5B-Instruct, local):** Un-staged slot-constrained decoding admitted all 40/40 proposals with 3/40 (7.5%) ExactCoboundary and 1/40 (2.5%) ℓ_1-optimal (39/40 suboptimal admissions). Under P* complete guard G_{complete}, all 40/40 failed Stage 1 after 1 retry and abstained via NullTree (0 suboptimal Stage-2 admissions).
- **Panel 2 (qwen/qwen-2.5-7b-instruct, OpenRouter, 121 calls at $0.018452):** Un-staged slot-constrained decoding admitted all 40/40 proposals with 9/40 (22.5%) ExactCoboundary and **0/40 (0.0%) ℓ_1-optimal**. Under P* complete guard G_{complete}, Attempt 1 produced 4 partial-guard passes ($G_{\text{partial}} = 1$), of which **3 were concrete Local Green / Global Red**

witnesses ($\hat{s} \neq \sum_j c_j$, Theorem 6) caught only by G_{complete}. After 1 diagnostic retry, 2/40 cases passed G_{complete} and were admitted to Stage 2, and **both (2/2 = 100.0% conditional Stage-2 precision)** achieved certified ExactCoboundary and exact ℓ_1-optimality (38/40 NullTree abstentions, 0 suboptimal admissions).

6.9 Trial 11 (C006-S8): Held-out 60-board multi-cycle campaign ($b_1 \in \{3, \ldots, 8\}$ across 7 graph families)

To test Theorem 8 across 60 held-out multi-cycle boards spanning cycle ranks $b_1 \in \{3, 4, 5, 6, 7, 8\}$ (10 boards per cycle rank across K4, K3,3, cube-Q3, K5, Petersen, wheel-7-rim, and wheel-8-rim), we executed c006_60board_hidden_p_star_receipt.json:

- **NO_LLM exact baseline (no_model_exact_c006_60board_baseline):** 60/60 (100.0%) ACCEPTED_ZERO_DEFECT ($\Phi_1 = 0$) and 60/60 zero-send replay_session verified across all 6 cycle ranks and 7 graph families.
- **Panel 1 (Qwen/Qwen2.5-0.5B-Instruct, local 6-board stratified anchor panel across $b_1 \in \{3..8\}$):** Un-staged 3-turn session solved 0/6 (0.0%); Two-Stage Tree-Then-Chord P* abstained at Stage 1a via NullTree on 6/6, preventing **24 wasted downstream turns** (0 corrupted Stage-2 admissions).
- **Panel 2 (qwen/qwen-2.5-7b-instruct, OpenRouter, all 60 hidden boards, 301 calls at $0.097765):** Un-staged 3-turn action generation solved **0/60 (0.0%)** across all cycle ranks $b_1 \in \{3..8\}$. Under Two-Stage Tree-Then-Chord P*, Stage 1a (G_T) passed on 8/60 boards (4/10 on K_4 at $b_1 = 3$, 1/10 on wheel-7-rim at $b_1 = 7$, and 3/10 on wheel-8-rim at $b_1 = 8$) and abstained via NullTree on 52/60, preventing **208 wasted downstream turns (52 \times 4)**. Stage 1b (G_C) passed on 7/8 on Attempt 1 and 8/8 (100.0%) after 1 retry. All 8/8 admitted boards (**100.0% conditional Stage-2 precision; 8/60 = 13.3% unconditional**) terminated in certified ACCEPTED_ZERO_DEFECT ($\Phi_1 = 0$) and passed zero-send replay_session.

6.10 Trial 12 (B-COH-S1): Multi-stage P* cohomology certificate verification across C011–C020

To test Theorem 12 across all ten cohomology challenges (C011–C020), we extended check_certificate in omega_engine_pfp/domain/cohomology/ to independently verify all 328 JSON benchmark cases (278/278 SOLVED, 36/36 OUT_OF_SCOPE, 14/14 UNKNOWN rejected) and executed the paired multi-stage P* campaign (b_cohomology_p_star_receipt.json):

- **NO_LLM exact baseline (no_model_exact_b_cohomology_baseline):** 10/10 (100.0%) SOLVED certificates verified + 10/10 (100.0%) OUT_OF_SCOPE certificates verified across C011–C020.
- **Panel 1 (Qwen/Qwen2.5-0.5B-Instruct, local):** Un-staged 0/10 verified; P* certificate-guarded caught 1 false SOLVED claim on Attempt 1 and abstained via NullTree on 10/10 (0 admitted false SOLVED verdicts).
- **Panel 2 (qwen/qwen-2.5-7b-instruct, OpenRouter, 30 calls at $0.020035):** On Attempt 1, qwen/qwen-2.5-7b-instruct emitted well-formed JSON with "status": "SOLVED" on **10/10 (100.0%)** of challenges C011–C020, every one of which violated the exact algebraic identities of its challenge (confirming Theorem 12.1). Independent check_certificate caught all 10/10 false SOLVED claims on Attempt 1, triggered 1 diagnostic retry, and projected all 10 persistent failures to NullTree (admitted_false_solved_count = 0, conditional_stage2_precision = 1.0).

7. Executable replay and limitations

The replay script uses only the Python standard library and executes twelve deterministic verification suites with zero model calls:

1. **Glyph code round trips and rejection controls:** Verifies all four Boolean code/assignm ent round trips (`<00>`, `<01>`, `<10>`, `<11>`), refuses malformed strings, and confirms that only `<01>` and `<10>` satisfy $x + y = 1$.
2. **Cochain primal-dual identity (Theorem 2):** Verifies $d_1 d_0 = 0$, $\Phi_1(c) = \Gamma_2(c) = |s|$, node-potential integration $d_0 a = c'$, and lightest-edge weighted ℓ_1 optimality across all 8 C004 cycle cases.
3. **Coloured operad `NullTree` absorption (Theorem 3):** Verifies that valid colour-matched grafts succeed while colour mismatches, out-of-bounds slot indices, and `NullTree` children or parents are absorbed to `NullTree` ($f \circ_i \mathbf{0} = \mathbf{0}$ and $\mathbf{0} \circ_i g = \mathbf{0}$).
4. **BPE prefix-automaton viability (Proposition 4):** Verifies that the integer-slot prefix DFA admits multi-character BPE boundary tokens (`' -'`), blocks premature `<eos>` before slot k, and accepts completed k-slot tuples.
5. **Single-cycle guarded Stage-1 retry and partial-guard false-green kernel (Theorems 5 & 6):** Verifies the exact bijection $\Delta s \mapsto z_1(\Delta s) \in \mathcal{K}_{\text{false}}(x)$ with Stage-2 residual $\Phi_{1,w}(c'(\Delta s)) = w_{i^*}|\Delta s| > 0$, confirms the `c004_k6_01` retry witness `(0, 2, 4, 4)` at $\Delta s = -3$, and checks that G_{complete} closes the false-green kernel to $\emptyset$.
6. **Multi-cycle K_4 triangular chord decoupling and rank-6 visibility gap (Theorem 7):** Verifies across all 4 K_4 multi-cycle overlap boards ($b_1 = 3$, shared tree edge incidence 3) that $G^{\text{C006}}_{\text{complete}}$ implies exact zero cochain defect ($c' = d_0 p^*_T$, $\Phi_1(c') = 0, \Gamma_2(c') = 0$), and verifies the 3 empirical multi-cycle Local Green / Global Red witnesses in $\mathcal{K}^{\text{C006}}_{\text{false}}(x) \cong \mathbb{Z}^6 \setminus \{0\}$.
7. **Higher-Betti ($b_1 \in \{4, \ldots, 8\}$) two-stage tree-then-chord slot-arity reduction and kernel closure (Theorem 8):** Verifies across $K_{3,3}, Q_3, K_5$, Petersen, and W_7 that two-stage factorization strictly reduces maximum integer-slot arity from $(|V| - 1) + 2b_1 \in \{13, 17, 16, 21, 21\}$ to $\max(|V| - 1, b_1) \in \{5, 7, 6, 9, 7\}$ and that $G_T \wedge G_C = 1 \iff (\hat{p}_T, \hat{c}'_{E \setminus T}) = (p^*_T, d_0|_{E \setminus T} p^*_T)$.
8. **Dichotomic Fredholm-alternative guard closure and false-verdict elimination (Theorem 9):** Verifies across all 8 public `C005` difference boards (`pub_cons_01..pub_cons_08`) that $G_{\text{Fredholm}} = 1$ uniquely determines $(\hat{p}_T, \hat{r}_{E \setminus T}) = (p^*_T, r^*_{E \setminus T})$ and rejects lazy false-`INCONSISTENT` and false-`CONSISTENT` perturbations.
9. **Cohomological syndrome decoding $\mathbf{r}^* = S\mathbf{c}$ across Verdict, Single-Edge Chord/Tree Repair, and Audit (Theorem 10):** Verifies on `C001` rank-difference graphs that $\mathbf{r}^* = S\mathbf{c}$ vanishes if and only if $\mathbf{c} \in \text{im}(d_0)$ and decodes both chord perturbations ($\mathbf{r}^* = -\hat{\delta}\mathbf{e}_m$) and spanning-tree perturbations ($\mathbf{r}^* = \hat{\delta} S_{\cdot, e^*}$, including c001_repair_1101 `['D','C'] -> 2408` across 4 chord cycles).
10. **Tree-coboundary defect migration vs. path-integrated B_T^{-1} guard diagnostics (Theorem 11):** Verifies on rooted spanning trees of depth ≥ 2 that an internal tree-edge integration error $\Delta_{v^*} \neq 0$ shifts $\text{Subtree}_T(v^*)$ while leaving descendant edge-local residuals zero, so updating only v^* migrates the coboundary defect to child edges ($v^* \to w$), whereas $\delta_T = B_T^{-1}\mathbf{e}_T$ flags the entire subtree $\text{Subtree}_T(v^*)$ simultaneously.
11. **Independent exact cohomology certificate verification (`C011`–`C020`, Theorem 12):** Verifies on finite `C015` ($\mathbb{F}_2$ persistence $R = D \cdot V$) and `C018` (signed-graph switching frustration) instances that status-label-only acceptance admits invalid `"status": "SOLVED"` payloads whereas independent algebraic recomputation rejects them.

12. **Offline recount of all 13 committed C001, C004, claude-fable-5-1, C005, C006, and `b_cohomology` receipts:** Replays all six committed C004 benchmark receipts (including the 40-case hidden ladder `C004-S6`), the accounted claude-fable-5-1 theorem-derivation receipt, all three C006 receipts (`C006-S6` K_4, `C006-S7` higher-Betti, and `C006-S8` 60-board hidden campaign), the `C005-S1` dichotomic Fredholm receipt, the `C001-S1` distractor-filtered syndrome receipt, and the `B-COH-S1` (`C011`–`C020`) cohomology certificate receipt from disk (when run inside the repository checkout) with zero network or provider calls.

Claim hygiene summary

1. **What this paper supports:**
   - Theorem 1 (channel-independent host checking soundness), Theorem 2 (single-cycle prompt-to-cochain ℓ_1 primal-dual coboundary identity), Theorem 3 (two-sided `NullTree` ideal absorption in $\mathcal{O}/\mathcal{I}$), Proposition 4 (BPE prefix-automaton logit-mask viability and `F-null` token identity), Theorem 5 (guarded Stage-1 slot retry admission rate, conditional precision, and Stage-2 call savings), Theorem 6 (single-cycle partial-guard visibility gap $\mathcal{K}_{\text{false}}(x) \cong \mathbb{Z} \setminus \{0\}$), Theorem 7 (multi-cycle triangular chord decoupling and rank-$(|V| - 1 + b_1)$ partial-guard visibility gap $\mathcal{K}^{\text{C006}}_{\text{false}}(x) \cong \mathbb{Z}^{|V|-1+b_1} \setminus \{0\}$), Theorem 8 (two-stage tree-then-chord `P*` slot-arity reduction from $|V| - 1 + 2b_1$ to $\max(|V| - 1, b_1)$ and two-stage false-green kernel closure), Theorem 9 (dichotomic Fredholm-alternative `P*` guard closure and elimination of the lazy false-refusal attractor), Theorem 10 (cohomological syndrome decoding $\mathbf{r}^* = S\mathbf{c}$ across `verdict`, chord/tree single-edge `repair`, and `audit`), Theorem 11 (tree-coboundary defect migration under coordinate-local retry vs. one-step subtree identification under path-integrated B_T^{-1} diagnostics), and Theorem 12 (independent exact certificate verification across `C011`–`C020` closing the status-label false-`SOLVED` kernel) under their declared finite assumptions.
   - Across all 12 empirical mechanism and cohomology trials (`C004-S1..S6`, `C006-S6..S8`, `C005-S1`, `C001-S1`, and `B-COH-S1` across `C011`–`C020`), `P*` staged slot materialization with complete deterministic Stage-1 guards and `NullTree` abstention achieved `40/40 (100.0%)` **conditional Stage-2 precision** (`0` admitted false `SOLVED` verdicts across `242` guarded board/challenge evaluations), caught `3` additional Local Green / Global Red witnesses on the 40-case hidden `C004` ladder and `10/10` false `SOLVED` status claims on `B-COH-S1`, and prevented `208` wasted downstream turns on the 60-board hidden `C006` campaign.
2. **What this paper does not support:**
   - Does not claim that Coloured Operad slot-grafting, Layer-2 grammar projection, or a single Stage-1 retry gives small or 7B models general symbolic competence: on the 40-case hidden `C004` ladder (`C004-S6`), qwen/qwen-2.5-7b-instruct passed `G_complete` on `2/40` (`5.0%`) cases; on the 60-board hidden `C006` campaign (`C006-S8`), it passed $G_T \wedge G_C$ on `8/60` (`13.3%`) boards; and on `B-COH-S1` (`C011`–`C020`), it passed check_certificate on `0/10` (`0.0%`) challenges (`10/10` abstained via `NullTree`).
   - Does not claim that local `15.24 GB` bf16 7B execution fits within `4.0 GB` laptop GPU VRAM without quantized GGUF/AWQ backends or remote provider routing.
3. **The narrowest defensible claim:**
   - Decomposing multi-constraint algebraic and cohomological tasks into bounded-arity `P*` Stage-1 slot-materialization cuts guarded by complete deterministic linear/coboundary, path-integrated B_T^{-1}, and independent certificate predicates ($G_{\text{complete}}, G^{\text{C006}}_{\text{complete}}, G_T \wedge G_C$, $G_{\text{Fredholm}}, G^{(m)}_{\text{cert}}$) with `NullTree` abstention enforces `100%` **conditional Stage-2 precision**

(40/40 across all admitted grafts in C004-S5, C004-S6, C006-S6, C006-S7, C006-S8, C005-S1, C001-S1, and B-COH-S1, with 0 admitted false verdicts) while converting unrecoverable model arithmetic or witness failures into zero-downstream-cost NullTree abstentions.

4. **The next test needed:**
   - Evaluate whether per-challenge multi-stage P* operator decomposition for C012 (sheaf coboundary matrix d_0 materialization before H^0/H^1 kernel/cokernel extraction) and C015 (column-by-column $\mathbb{F}_2$ pivot reduction $R = D \cdot V$) lifts 7B/frontier verified solve rates above 0/10 under the same fail-closed check_certificate gate.

8. Sources and relation to the programme

This draft builds on the WDP manuscript, Paper 021 (source and execution audit), Corrected challenge findings, Paper 022 (frontier research), the Categorical Operad Prompt Architecture, and the bounded C001/C004/C005/C006/b_cohomology mechanism sessions. The two-level teaching packet is retained as an unexecuted structural example. Genesis and MSS-N presentation stages do not license a new mathematical or physical carrier.

Research extension: evolving the proposal harness

Prompt, workflow, search-policy and context-policy rewrites are additional proposal-channel interventions. The host's task and checking obligations remain fixed. A strategy label, root-cause explanation or compressed context can change the distribution of submitted candidates without changing what a valid certificate must satisfy. This is the operational use of channel-independent checking; it does not establish a theorem about empirical improvement.

The new campaign uses named Anthropic deployments with frozen role configurations. Its interface supplies adaptive-thinking and output limits; it does not expose provider seeds, arbitrary logit interventions or attention weights. Local seeds reproduce scheduling choices only. Every paid attempt, including an unusable response, is bound to the original budget and retained raw evidence. Changed candidate code is executed only within the declared isolated surface.

The historical mechanism trials and finite proofs above retain their own source and evidence identities. They are not training data or measured outcomes for the new held-out panel. Paper 029 supplies the separate protocol, ablations and exploratory ignition design. Those new model effects remain NOT_MEASURED pending the verified remote host.

The source and page references for this extension are collected in Paper 029 and its source-to-claim map. The author-review rendering does not replace a historical deposit or archive.